

Providing Liquidity in an Illiquid Market: Dealer Behavior in U.S. Corporate Bonds

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May 2018

Abstract

We examine market making behavior of dealers for 55,988 corporate bonds, many of which trade infrequently. Dealers have a substantially higher propensity to offset trades within the same day rather than committing capital for longer periods for riskier and less actively traded bonds. Dealers' holding periods do not decline with a bond's prior trading activity, and in fact are lowest for some of the least active bonds. As a result, cross sectional estimates of roundtrip trading costs do not increase as prior trading activity declines. Our results suggest that dealers endogenously adjust their behavior to mitigate inventory risk from trading in illiquid and higher risk securities, balancing search and inventory costs in equilibrium such that observed spreads can appear invariant to expected liquidity.

We thank Terrence Hendershott, Charles Himmelberg, Stacey Jacobsen, Norman Schürhoff, participants at the 2015 FINRA-Columbia Fixed Income Conference, 2016 Banque de France and Toulouse School of Economics (TSE) conference on Market Liquidity and Regulation, 2017 Fixed Income and Financial Institutions Conference, and seminar participants at Boston College, University of Illinois Urbana Champaign, University of Nebraska, The Queen's University (Belfast), U. Sydney, U. South Australia, U. Auckland, U. Melbourne, Queensland University of Technology, Bond University, Macquarie University, University of Technology-Sydney, and Australia National University.

I. Introduction

Despite the vast number and extremely large dollar volume of U.S. corporate bond issues, many corporate bonds never or rarely trade. While the importance of illiquidity has been extensively studied with regard to corporate bond pricing, the difficulty of market makers in providing liquidity to this over-the-counter market is less well understood. For illiquid bonds, dealers may be unwilling to commit capital in a market where they face both significant inventory risk upon purchasing an asset from a customer and potentially high search costs in locating a counterparty for an offsetting trade.

In this paper, we examine the market making behavior of dealers for 55,988 U.S. corporate bonds, with a focus on the majority of bonds which are infrequently traded and expose dealers to significant inventory risk. We show that dealers' behavior endogenously changes based on the riskiness and illiquidity of the assets they trade. Whether dealers take bonds into inventory depends on factors that drive expected search and inventory costs, including bond characteristics, dealer and counterparty characteristics, market conditions, and the competitive structure of the dealer market for a given bond. We find that dealers are most likely to offset trades quickly, rather than holding bonds in inventory overnight or longer, for the least actively traded and riskiest bonds. We further show that the choice to provide liquidity is reflected in observed spreads on dealers' trades. Because dealers can mitigate inventory risk by more actively searching for a counterparty, observed spreads are often lower for riskier bonds and those with little or no prior trading activity.

Our work draws on a growing theoretical literature that describes trading behavior in over the counter (OTC) dealer markets. Starting from Duffie, Gârleanu, and Pedersen (2005), these models show that active search for a counterparty arises endogenously, either by investors looking for competing market makers or by market makers looking for an offsetting trade. The intensity of search affects prices, which are determined by a bilateral bargaining process that reflects the investor's and dealer's alternatives to trading

immediately. Higher search costs imply worse outside alternatives and worse prices; for example, investors that do not have easy access to alternative market makers sell to the market maker at a larger discount.¹

Inventory models provide a different view of dealers' role when there is uncertainty about the arrival of buyers and sellers - dealers provide liquidity by committing capital when there is an imbalance (Garman, 1976; Stoll, 1978; Amihud and Mendelson, 1980). Ho and Stoll (1981) show that additional uncertainty about the return on assets held in inventory affects dealers' willingness to act as a market maker as well as the bid ask spread which compensates for this risk. The empirical implication is that observed spreads are higher when there is a greater risk of price movements during the dealer's holding period and when trades are not expected to be quickly offset. Further, dealers willingness and/or ability to commit capital can be limited (Hendershott and Seasholes, 2007; Brunnermeier and Pedersen, 2009; Duffie, 2010).

Combining the arguments of Duffie, Gârleanu, and Pedersen (2007) and other search models with the realities of dealers' capital costs and inventory risk, dealers will offer customers a tradeoff between pricing and immediacy (liquidity). However, when inventory risks are sufficiently high, we expect dealers' search to intensify, such that dealers act more as brokers by pre-arranging matched buy and sell trades.² Dealers provide little immediacy when there are few trading opportunities, for example for a bond that trades at best once a month; investors retain price risk while dealers search for a counterparty to offset their trade, as well as the risk that their trade is not ultimately executed. Unlike inventory models, this implies that observed spreads can be lower for bonds with greater potential inventory risk when dealers mitigate this risk by searching for counterparties. Our empirical results support these conjectures, particularly for the least actively traded and riskiest bonds.

As Schestag, Schuster, and Uhrig-Homburg (2016) note, prior corporate bond research often utilizes liquidity measures based on those previously used in equity markets. While immediacy can be measured in

¹ In Duffie, Gârleanu, and Pedersen (2005), market makers are assumed to off-load inventories in a frictionless interdealer market. Duffie, Gârleanu, and Pedersen (2007) do not model market makers, but do incorporate "holding costs" of an investor counterparty, which implies that dealers buy from investors at lower prices when there are fewer alternative buyers, price volatility is higher, or risk aversion of the buyer is higher.

² Harris (2015) and Schultz (2017) note that rather than committing capital, dealers will search for a counterparty prior to executing a difficult to offset trade, for example for large trades or speculative bonds. As described below in Section III.C, the incidence of seemingly pre-arranged trades in corporate bonds has been noted by several researchers.

milliseconds or microseconds for foreign exchange or equities (Chen, Foley, Goldstein, and Ruf, 2017), the concept of immediacy for many bonds can be measured in hours, if not days or weeks. Our empirical analysis first examines measures of trading activity in order to develop proxies for the expected arrival time for a counterparty at the time of a dealer purchase. Observations of transaction prices for many bonds are rare: the median bond in our sample has only 0.6 institutional sized trades per month, and many bonds do not trade at all. However, many of the liquidity measures previously applied to the bond market require multiple trades in the same day, thereby excluding relatively inactive bonds. In contrast, we intentionally retain less active bonds in our sample and assign all bonds to trading activity deciles based on their trade count and number of non-zero trading days in the 30 day window preceding any dealer purchase.³ We show that these measures are stable over monthly or yearly periods, such that recent trading activity provides a good proxy for expected near term activity.

Our dataset allows us to identify purchases and sales reported to TRACE by individual (anonymous) dealers, enabling us to identify 8,576,027 roundtrips in which we match a dealer buy transaction to one or more offsetting sell transactions of that dealer in the same bond.⁴ For each roundtrip, we observe how quickly the buy transaction is offset (the dealer's holding period), the type of counterparties involved (dealer versus customer), and the roundtrip cost (spread) for the dealer trading the bond. Our analysis is based on roundtrips starting with a buy trade of at least 100 bonds, as larger trades are important in understanding dealers' inventory risk.

For roundtrips overall, a surprisingly large portion (58.2%) of dealer roundtrips are completed within the same day. Over 70% of these daily roundtrips are cases of a "paired" roundtrip where the dealer offsets the buy transaction with a single sell transaction of the same quantity, considerably higher than previously

³ Both trade count and days traded are related to the (in)frequency of price observations, but do not require multiple transactions per day and thus are appropriate for less actively traded bonds, unlike other commonly used measures. For example, calculating the Amihud (2002) measure of price impact or a measure based on Roll (1984) requires multiple trades per day. For further discussion of problems in applying traditional liquidity measures to infrequently traded bonds, see Chen, Lesmond, and Wei (2007), Nashikkar, Subrahmanyam, and Mahanti (2011), and Section III.A of this paper.

⁴ Dealer roundtrips match dealer purchases of a bond with one or more offsetting sell trades, closest in time, in the same bond to a customer or another dealer. Details of our methodology for identifying roundtrips are provided in Section III.C.

reported for institutional sized trades (Harris; 2015; Zitzewitz, 2010). For the remaining 41.8% of roundtrips, dealers take bonds into inventory at least overnight, with a median holding period lasting 28 days. Unless the dealer actively searches for a counterparty, bonds with few or no trades in the recent past expose the dealer to greater expected inventory risk. Inventory risk further increases for lower credit quality bonds with greater risk of large price movements. Comparing the lowest rated, least actively traded bonds to the highest rated most active bonds, dealers offset trades within the same day for approximately 75% versus less than 55% of roundtrips, respectively.

The dealer's roundtrip spread is calculated as the difference in the dealer's buy price and weighted average sell price. Since the effect of trading activity on dealer behavior may be non-linear in credit quality and quite different around the investment grade/non-investment grade cut-off, we control for credit rating by separately examining four credit rating groups (A and above, BBB, BB, and B and below). Our initial univariate results show that spreads do not monotonically increase for lower trading activity deciles relative to more actively traded bonds, and for the lowest quality bonds appear invariant to prior trading activity. We also observe that spreads are lowest when the dealer both purchases and sells in the interdealer market, and are notably higher if the dealer sells to a customer, indicating the value of locating a customer.

Thus, observed spreads will depend on dealers' decision whether to take bonds into inventory. This setting fits naturally within an endogenous switching regression framework.⁵ A first stage probit is used to predict whether dealer roundtrips are completed within the same day versus over longer holding periods, and the second stage model is used to explain roundtrip spreads. To understand the relationship between prior trading activity and dealers' behavior, the first stage model includes dummies for each of the five lowest trading activity deciles. Notably, for all credit qualities, bonds in the lowest decile of trading activity prior to the roundtrip have a significantly greater likelihood of roundtrips completed the same day relative to more active bonds. This result suggests that despite (or due to) the low level of trading in these bonds, dealers actively search for and find same-day counterparties. For example, bonds rated A or above in the lowest

⁵ See Hendershott and Madhavan (2015), Bessembinder and Venkataraman (2004), and Madhavan and Cheng (1997) for discussion of these models.

decile of days traded prior to the start of the roundtrip have a 12% higher likelihood that the roundtrip is completed the same day. The coefficients decline as trading activity increases (within the bottom deciles), remaining strongly positive and significant for the days traded measure and even becoming negative based on trade count. Overall, the first stage models show a strong relationship between our proxies for expected near term liquidity and the probability that dealers take bonds into inventory at least overnight. These results are consistent with stronger dealer search precisely where it is needed most, i.e., for the least active bonds.

Our first stage probit further includes bond and roundtrip trade characteristics related to expected price (inventory) risk and dealers' search. Bond characteristics that indicate fewer or better informed potential counterparties, such as older bonds and Rule 144A private placement bonds, have a significantly higher probability of roundtrips being completed the same day. Roundtrip trade characteristics include the size of the buy trade and identity of the counterparty which are also expected to affect probabilities: if larger trades take longer to unwind and expose the dealer to greater inventory risk, we would expect roundtrips starting from a larger dealer purchase to more likely be completed the same day. We find that large block-size purchases of lower credit bonds are more likely offset within the same day, and that the magnitude of this effect is quite large for lower credit quality bonds. We also find that purchases from another dealer have a substantially lower probability of being held in inventory. This suggests that rather than assuming inventory risk from another dealer, these dealers are compensated for locating a customer to offset the trade, similar in spirit to Duffie, Gârleanu, and Pedersen (2005).

At the same time, more active dealers, measured either by their total reported trade count or volume, may be better able to locate counterparties at lower cost. Despite the infrequent trading of many bonds, a large number of dealers report trades to TRACE. Consistent with models of a core-periphery network in OTC markets (Neklyudov, 2014; Hollifield, Neklyudov, and Spatt, 2017), market making is highly concentrated; on average in our sample, 38% of a bond's trading volume is done by the top ten dealers as ranked by dealers' total trading volume. We find that dealers with a large share (top ten) of the total market volume provide much of the market's liquidity by taking bonds into inventory. In contrast, dealers with a large share of total market trade count have a higher likelihood of completing roundtrips in the same day.

The latter group includes many smaller dealers which execute relatively more small trades. We further find a strong effect that dealers that are more central in the interdealer network are more likely to take bonds into inventory. Perhaps not surprisingly, dealers that have taken the same bond into inventory in the past, or have traded the same bond more frequently in the prior year, are significantly more likely to take the bond into inventory overnight. These effects continue to hold when we exclude the top ten largest dealers by trade volume from our sample.

The second stage of the endogenous switching regression models estimates two equations explaining roundtrip spreads, one equation for roundtrips completed the same day and one for those completed over longer periods. Unlike results in equity markets where larger blocks are more expensive to trade, spreads are significantly lower for roundtrips starting from larger dealer buys, regardless of whether roundtrips are completed in the same day or longer. Spreads are also higher when the dealer transforms a larger buy by breaking up the trade into smaller sized sells, consistent with Edwards, Harris, and Piwowar (2007), Goldstein, Hotchkiss, and Sirri (2007), and others who document higher spreads for smaller trades.

Second stage model results for the trading activity deciles show that, except for bonds in the lowest deciles and the highest credit quality bonds, spreads for infrequently traded bonds are the same or smaller than the roundtrips spreads for more active bonds. In addition, these regressions show that the magnitude of the effect of expected liquidity is greater for roundtrips completed over more than one day. Excluding roundtrips of top ten dealers, we find that the effect of expected liquidity on spreads is even greater for bonds in the least active trading decile. Together with our first stage selection model, our results show that the choice of whether or not to take the bond into inventory overnight (versus completing the roundtrip within the day) is important in understanding observed spreads when many bonds trade infrequently.

We also use our second stage estimates to calculate a counterfactual; using roundtrips completed the *same day*, we calculate their predicted spread using the coefficients estimated from roundtrips completed in *more than one day*. For example, using deciles either by trade count or days traded, we estimate that roundtrip costs would be 13 basis points higher for bonds rated A and above but as high as 25 basis points higher for BBB rated bonds had they not been completed the same day. Similarly, spreads on roundtrips

completed in more than one day for BBB bonds are estimated to be 9 basis points higher than they would be if completed on the same day. Of course, these estimates are based on dealers' arguably optimal decision as to whether to take bonds into inventory, as well as whether to trade at all. Regardless of magnitude, the sign of these estimates is indicative of the tradeoffs in price when dealers provide immediacy, and is consistent with dealers undergoing search to offload inventory quickly and being compensated for retaining bonds in inventory overnight or longer.

The fact that dealer behavior varies with trading activity is particularly important in understanding the behavior of spreads and other measures of dealer activity in periods of large increases in overall market illiquidity. Prior research (Bao, Pan, and Wang, 2011; Dick-Nielsen, Feldhütter, and Lando, 2012; Friewald, Jankowitsch, Subrahmanyam, 2012) has shown large increases in measures of illiquidity for corporate bonds during the financial crisis. We find that dealers are significantly more likely to complete roundtrips within the same day in this period (defined as the second quarter of 2007 through the second quarter of 2009) only for investment grade bonds. These findings suggest that dealers engage in more search and provide less immediacy to this portion of the market during the crisis, consistent with the costs of limited capital at this time. We further observe higher spreads for roundtrips of high credit quality but lower trading activity decile bonds, but only when bonds are taken into overnight inventory. We find similar results when we examine days of large increases in the VIX index, which have also been shown to experience increases in bond market illiquidity (Bao, Pan, and Wang, 2011; Anderson and Stulz, 2017). These results show that in times of market stress, dealers are more likely to quickly offset trades even for higher rated bonds that dealers more commonly take into inventory, and that there is a greater effect of illiquidity on spreads for these bonds.

Our paper contributes to several strands of recent literature that study trading in the over-the-counter U.S. corporate bond market. First, our results provide insight into the ongoing debate on the state of corporate bond market liquidity. This literature has produced two seemingly contradictory conclusions: one that liquidity has improved based on observed declines in transactions costs, and the other that dealer capital commitment has declined over time (Adrian, Fleming, Shachar, and Vogt, 2017; Bessembinder et al, 2017; Choi and Huh, 2017; and Bao, O'Hara, and Zhou, 2018). Our work is empirically consistent with both

conclusions, but points to the explanation that dealers take on less risk in observed trades as expected liquidity declines. Further, our paper provides new insights related to a growing number of studies showing that typical measures of corporate bond liquidity decrease sharply during the financial crisis and more generally in times of market stress (Dick-Nielsen and Rossi, 2017; Trebbi and Xiao, 2017; and Anderson and Stulz, 2017). Our results show that in periods when even when more actively traded higher credit quality bonds become illiquid, dealer behavior becomes more similar to that observed for the larger number of riskier and less actively traded issues.

Second, it is important that our study incorporates the vast number of corporate bonds that trade infrequently, given this segment of the market receives relatively little attention in much of the research in this area (see discussion in Section III.A. of this paper). Applying methodologies to measure liquidity that have previously been used on higher frequency data can be either infeasible or lead to different inferences when attempting to apply them to less actively traded corporate bonds. Importantly, we observe striking differences in dealers' behavior based on the prior trading activity of a given bond, which is key to understanding our novel result that observed spreads on inactive bonds are similar to the most active bonds. Most broadly, recognizing that dealers' endogenously adjust their behavior based on the liquidity of the security they trade is important in interpreting any studies that seek to explain observed measures of liquidity either cross-sectionally or over time. Our results also speak to attempts to measure execution costs for corporate bonds, since comparing transactions prices to a prior trade price or benchmark is particularly problematic when bonds have previously traded once a month if at all.⁶

Third, our paper contributes to recent work that considers inventory behavior of dealers in the OTC corporate bond market. Friewald and Nagler (2016) show dealers' inventory is mean reverting and document a strong negative relationship between changes in inventories and risk adjusted bond returns, which complements our interpretation that dealers use pricing to control exposure to inventory risk. Anand, Jotikasthira, and Venkataraman (2017) demonstrate cycles in dealers' aggregate bond inventory, also

⁶ O'Hara, Wang, and Zhou (2018) examine the differential execution costs incurred for insurance company trades in corporate bonds and find that less active investors receive significantly worse execution. See also Greenwich Associates (2015).

consistent with inventory models where dealers adjust prices in response to large inventory imbalances. In contrast, our paper focuses on dealer's decision to provide liquidity by taking a particular bond into inventory at least overnight, and the simultaneously-determined spreads. As a result, we can explain behavior for less actively traded and riskier bonds that otherwise appears contrary to the predictions of more standard inventory models (for example, Amihud and Mendelson, 1980, and Ho and Stoll, 1981) that suggest spreads increase as trading decreases.

Finally, our paper is also related to several others including Zitzewitz (2010), Ederington, Guan, and Yadav (2014), Harris (2015), Choi and Huh (2017), Bessembinder et al. (2017), and Schultz (2017), which point out that dealers frequently purchase and sell the same bonds within one minute, most often offsetting a purchase from a customer by selling into the interdealer market. They interpret these essentially riskless trades as prearranged by dealers, and further show their incidence has recently increased. While these near-simultaneous matched trades account for a large portion of the "paired roundtrips" we identify, we look more broadly at trading within the day, the decision to take bonds into inventory overnight, and the relationship between observed spreads and the inventory decision. Consistent with models of dealer behavior in OTC markets, dealer behavior is strongly related to the type of counterparty (dealer versus customer) and characteristics of the dealer itself. Our results that trades with customers are more valuable to the dealer support the theoretical predictions of Condorelli, Galeotti, and Renou (2017) and the findings in Sirri (2014) for municipal bonds. Most importantly, we show that dealer decisions are significantly related to proxies for greater expected inventory risk, such as the prior trading activity of the bond, as well as other bond characteristics and market conditions which exacerbate inventory costs and risks.

In summary, our analysis demonstrates dealers' ability to adjust their market making activity based on the expected liquidity of the bond traded. As a result of this behavior, low credit quality bonds which rarely trade can be observed to have lower roundtrip costs than a highly rated active bond for which a market maker chooses to maintain inventory. Dealers mitigate inventory risk by acting more as brokers and less as

market makers as liquidity declines, but in doing so provide less immediacy to customers looking to trade less active bond issues.⁷

Our paper is organized as follows. Section II provides the theoretical framework for our hypotheses and analysis. Section III reviews liquidity measures used for corporate bonds, and describes our data and measures of dealer roundtrips. Section IV examines dealers' inventory behavior as well as how the spreads dealers charge on roundtrip trades vary with prior trading activity and credit risk. Sections V and VI estimate our model of dealer spreads that accounts for the endogeneity of dealers decision to take bonds into inventory, while Section VII implements this model to examine periods of market stress and further issues. Section VII concludes.

II. Theoretical framework and hypotheses

In this section, we review the theoretical framework relevant to OTC markets and specifically corporate bonds. We then derive our empirical implications.

II.A. Models of trading in OTC markets

It is difficult to provide a single unified framework to describe trading behavior in OTC markets. Our analysis bridges several strands of literature, including search models, models of endogenous intermediation and network formation, and earlier models of inventory behavior.

Search models of OTC markets provide our starting point. The seminal work of Duffie, Gârleanu, and Pedersen (hereafter DGP) studies the implications of search frictions for market making and pricing in an OTC market. In DGP's 2005 model, investors meet market makers with a fixed intensity, representing the sum of intensities of investors' search for market makers and market makers' search for investors. Investors sequentially bargain with market makers as they are found, and prices reflect the investor and dealer's

⁷ The problem of providing liquidity in a market with a large number of assets, many of which rarely trade, is similar to that found in the agency mortgage-backed securities market, where tens of thousands of unique securities have been issued (Gao, Schultz, and Song, 2017).

alternatives to trading immediately. Market makers are assumed to off-load inventories in a frictionless interdealer market.

In such models, higher search costs imply worse outside alternatives and worse prices. For example, a “sophisticated” investor that has access to alternative market makers can sell to (buy from) the market maker at a higher (lower) price. Notably, models of adverse selection predict the opposite relationship between investor sophistication and price. DGP (2007) further show that search frictions affect the resiliency of prices when outside opportunities and therefore bargaining positions deteriorate, i.e. search time increases.

A number of recent papers build on this framework and link search theory, with a centralized interdealer market, to network theory. In these papers, as in DGP (2005, 2007), investors contact dealers at a constant exogenous rate and trade at a price determined by bilateral bargaining. However, these papers model various types of heterogeneity in investors and/or dealers to capture other important features of OTC markets. For example, in Neklyudov (2014) and Hollifield, Neklyudov, and Spatt (2017), an interdealer network arises endogenously as a result of heterogeneity in dealers’ trade execution technologies; dealers with higher “speed,” which we interpret as the ability locate a counterparty, emerge as more interconnected core dealers.⁸ Execution technology, and therefore the dealer’s position in the network, can be altered with large investments but is fixed in the short run. Thus, prices are still determined by bilateral bargaining between randomly matched but connected participants, and reflect alternative trading opportunities. Chang and Zhang (2016) show that a core/periphery network structure arises based on differences in risk sharing, where institutions with the lowest risk sharing needs endogenously specialize as intermediaries and become the core.⁹

Other recent contributions demonstrate the chain of intermediation expected to occur in OTC markets. Hugonnier, Lester and Weill (2016) show that heterogeneity in investor valuations magnifies the

⁸ Lester, Rocheteau, Weill (2015) show the impact of search frictions when dealers are limited in their capacity to process multiple orders at once. Their model produces a trade-off between bid-ask spreads and execution speed.

⁹ Gofman (2011) models the outcome of a bilateral bargaining process where trade prices are affected by dealers’ private asset values. Other models based on random search where counterparties arrive at an exogenous rate include Lagos and Rocheteau (2009) and Afonso & Lagos (2015). The effects of an exogenously specified network structure are examined by Babus and Kondor (2016) and Malamud & Rostek (2012).

price impact of search frictions. In their model, assets move from natural sellers to natural buyers through a chain of intermediaries. Similarly, Condorelli, Galeotti, and Renou (2017) provide a dynamic model of bargaining in networks that introduces asymmetric information, which in the case of the OTC bond market may be information about a counterparty's liquidity shock or customer order flow. They show that an asset may travel through several layers in an interdealer network at declining prices, with the exception of the higher valuation final buyer in the chain.

Inventory models are also based on uncertainty about the arrival of buyers and sellers. However, in these models, dealers provide immediacy to less patient investors by committing capital when there is an imbalance (Garman, 1976; Stoll, 1978; Amihud and Mendelson, 1980; Ho and Stoll, 1981). Bid ask spreads compensate dealers for inventory risk, and uncertainty about the return on assets held in inventory affects both compensation for this risk and dealers' willingness to act as a market maker (see also Grossman and Miller, 1988, and Whitcomb, 1988).

Models of inventory behavior also show that dealers can manage positions by adjusting prices to attract orders, leading to mean reversion in observed inventories. Alternatively, dealers can unwind inventory in the interdealer market (Ho and Stoll, 1983). While interdealer trading enables inventory risk sharing, the initiating dealer must give up some portion of the spread to its counterparty. In these models, the benefits of risk sharing via the interdealer market outweigh the costs when dealers are farther from their desired inventory level. These models do not consider search costs, however, and so do not include the possibility that the dealer does a more active search instead of using the interdealer market. Further, dealers will vary in their willingness or ability to commit capital based on their inventory-carrying or risk-bearing capacity (Hendershott and Seasholes, 2007; Brunnermeier and Pedersen, 2009; Duffie, 2010).

II.B. Testable hypotheses

Customers in the OTC corporate bond market trade almost entirely through dealers as intermediaries. From the models described above, customers with worse outside trading opportunities face higher trading costs. Dealers provide liquidity (immediacy) to customers by purchasing a bond and taking it into inventory;

the dealer then waits or searches for a buyer, either locating a customer or selling in the interdealer market. The dealer's cost of providing immediacy is lower when it is better connected or is more active in the market for the bond, lowering search costs and the expected time to locate a counterparty. Costs are also lower as expected inventory costs decline, based either on characteristics of the bond or the dealer's own funding costs. A key implication of the search based models is that dealers offer, and customers face, a tradeoff between price and immediacy.

When the dealer's cost of providing liquidity is sufficiently great, two outcomes become more likely. The first is that a trade cannot be executed at a price acceptable to the customer. The second is that the trade occurs, but, the dealer acts as a broker and executes a pair of trades once the counterparty has been arranged. We refer to this outcome as a prearranged or "paired" roundtrip, the likelihood of which is increasing in the dealer's expected search costs and inventory costs. In such cases, dealers should still be compensated for searching for the counterparty, but, the customer meantime bears the holding costs as well as the risk the trade will not be executed at all.

We summarize our hypotheses as follows:

- *Trading costs are higher when investors have more limited trading options, and for bonds or in market conditions where dealers have higher expected search costs or inventory costs.*
- *Dealers are more likely to pre-arrange trades when their expected inventory costs are high. Trading costs are lower for roundtrips where dealers do not take on inventory risk, but, still increase with search costs.*

These empirical predictions differ from those often assumed based on inventory models. Instead, when dealers can endogenously change their behavior such that inventory risk is retained by customers, bonds with higher credit risk or which trade infrequently may not have higher observed trading costs in comparison to lower risk bonds for which dealers act as active market makers.

Models of endogenous intermediation also show that a larger share of the gains from trade accrue to the dealer that identifies a customer at the end of an intermediation chain (Condorelli, Galeotti, and Renou; 2017). When dealers purchase bonds from or sell bonds to other dealers, the more connected (core) dealer

likely captures the gains from trade either because of their connections or because they assume inventory risk. This implies the following:

- *Interdealer trades are likely to be pre-arranged, either because they enable the selling dealer to mitigate inventory risk or because the buying dealer has or can more easily locate a customer. Dealer roundtrip spreads are higher when dealers sell to a customer rather than in the interdealer market.*

III. Data and summary statistics for liquidity

This study uses data provided by the Financial Industry Regulatory Authority (FINRA) for all corporate bonds reported to TRACE from July 8, 2002 through March 31, 2011. In this section, we first discuss measures of corporate bond liquidity and define our proxies for expected near term liquidity. We then explain our sample construction and provide description of trading activity and dealer roundtrips.

III.A. Measures of liquidity for corporate bonds

As noted by Nashikkar, Subrahmanyam, and Mahanti (2011, p.628), “the absence of frequent trades in corporate bonds makes it difficult to compute trade-based measures of liquidity relying on quoted/traded prices or yields to measure liquidity, as has been done in the equity markets.” Further, many market microstructure models of dealer behavior, such as Kyle (1985) or Glosten and Milgrom (1985), do not explicitly assume frequent trades but do have implicit assumptions about the competitive nature of the market and the frequency of new information arrival, which implies more frequent trading.

Prior studies of corporate bond markets implement a number of different liquidity measures, largely based on those used for equity markets. The samples studied often *de facto* include only the most frequently traded bonds because different measures either require multiple trades close in time for the same bond, or use a regression approach for a broader set of bonds which still filters out the least actively traded bonds.¹⁰ For

¹⁰ Schestag, Schuster, and Uhrig-Homburg (2016) compare commonly employed liquidity measures and show that six different methods for estimating transactions costs provide similar estimates and are highly correlated. Their analysis is based on the 3,495 most actively traded bonds on TRACE which each trade on at least 75% of trading days during their life span. Estimates of transactions costs using a regression approach, as in Schultz (2001), Edwards, Harris, and Piwowar (2007), Bessembinder et al (2017), and others, may underweight bonds with fewer transactions or miss non-linearities in the data.

example, implementing an Amihud (2002) measure of price impact as in Dick-Nielsen, Feldhütter, and Lando (2012) requires at least two trades in the same bond in the same day. Using a daily Roll (1984) or similar measure as in Bao, Pan, and Wang (2011) requires two or three trades in the same bond in the same day. As described in Section III.B, our objective is to study dealer behavior in a market where many bonds trade only once or twice in a given month, if at all, as well as in times where even more frequently traded bonds experience a decline in liquidity, which requires that we specifically examine illiquid and infrequently traded bonds.

III.B. Sample and trading activity summary

Our initial sample includes 61,857 non-convertible bonds which trade at least once during our sample period and which can be matched with Mergent data for bond characteristics. We exclude bonds that do not trade at least once starting 60 calendar days after the bond offering date, bonds that do not have at least one dealer roundtrip (as defined below), and bonds that have no trades of at least 100 bonds, leaving a final sample of 55,988 bonds we use in our analyses. Table 1, Panel A shows that these bonds on average have a remaining maturity over five years, are over four years past their issuance date, and consist of 11% non-registered Rule 144A bonds. The median bond is A rated, while the mean bond rating is approximately BBB rated (credit rating of 3.8, where 1=AAA, 2=AA, etc.; see Appendix A1 for variable definitions).

We first demonstrate how actively – or inactively – corporate bonds trade. We calculate monthly trading statistics for each bond, assigning zeros to months where bonds do not trade, and excluding periods within 60 days of issuance, redemption, or maturity. Trade count is an important measure when many bonds have little trading activity. From Panel B of Table 1, the median bond trades 3.1 times per month, and institutional sized trades of 100 or more bonds occur only 0.6 times per month. The means are notably higher at 20 trades per month, 6.3 of which are institutional sized, showing the substantial variation in trading frequency. Trading volume is low: the median par amount traded is 717 bonds (\$717,000). We also report the number of non-zero trading days within the month (median of 1.3 days), since Dick-Nielsen, Feldhütter and Lando (2012) and others suggests this measure is priced in observed bond yields. Panel B further shows

that bonds on average trade less than half of the months eligible to be reported to TRACE. Our empirical tests divide bonds into deciles based on trading activity, exploiting the large variation in trading activity across bonds.

Bonds also vary in their concentration of trading by dealer. Despite their relative illiquidity, Panel C of Table 1 shows that bonds are traded on average by 55 different dealers over the full sample period.¹¹ Dealers also actively trade with each other, with an average of 28.0% of trades occurring in the interdealer market. To measure characteristics of the dealers trading a given bond, we identify the top ten and top 1% most active dealers as reported to TRACE by ranking each dealer by their total trade count across all bonds. Similarly, we rank and identify top dealers by trade volume. These rankings do not largely overlap, as dealers with smaller volume and large trade count likely focus their business on retail trading: of the top ten dealers ranked by trading volume, only three of those are also top ten dealers based on trade count. We then tabulate the proportion of each bond's trading by these top dealers. For example, on average 38.2% of a bond's trading volume is done by the top ten dealers as ranked by dealer trade volume. These statistics show the tremendous concentration of trading by the most active dealers, consistent with recent descriptions of core-periphery networks in OTC bond markets by Hollifield, Neklyudov, and Spatt (2017), Li and Schürhoff (2015), and others. Our finding that the top 1% of dealers account for about two-thirds of all trading by bond is similar to the finding in Di Maggio, Kermani, and Song (2017) that 50 dealers account for about 80 percent of all transactions.

Given the large variation in trading, we report the distribution of our trading activity measures (trade count and number of days traded) by deciles in Table 2, where decile 1 is the lowest frequency trading group.¹² Since we are primarily interested in less actively traded bonds, we report the lower five deciles individually but combine the most active five deciles (deciles 6-10) into one group, which therefore includes all bonds more active than the median bond, which is itself infrequently traded. Given the low

¹¹ The anonymous dealer identifiers are grouped by FINRA so that entities trading with multiple identifiers, such as large broker dealers, are considered a single dealer. After such grouping, there are 2,984 dealers in our dataset.

¹² We do not use volume to define trading activity deciles for our analysis. Prior literature shows that volume as a measure of liquidity behaves differently than other measures (Johnson, 2008; Anderson and Stulz, 2017) and is not priced in corporate bonds (Dick-Nielsen et al., 2012).

trading frequency of even the median bond, the deciles below the median contain highly illiquid bonds.¹³

Results in this paper are qualitatively similar to defining trade count deciles using only trades of greater than 100 bonds.

Both trading activity measures show wide variation in trading frequency from the most to the least active bonds. For example, bonds in the lowest decile of trade count have a median trade count of 0.1 trades per month, versus 12.3 trades per month for the more active group (deciles 6-10), which itself has on average less than a trade every other day. Similarly, days traded per month is 0.1 days for the lowest decile and 4.5 days for the more active group (deciles 6-10), further showing that the lowest deciles include highly illiquid bonds. Though not reported for brevity, we also find that bonds in deciles 6-10 trade on average in 82% of sample period months, while the most infrequently traded bonds in decile 1 trade in only 5% of the sample period months. Interestingly, controlling for trading activity, we do not observe striking differences across rating categories.

These trading measures are stable through time: Appendix Table A2 shows the stability of trading activity deciles on both a yearly and monthly basis. Depending on the trading activity measure, approximately 85% of bonds in the lowest activity decile 1 remain so in the following year, and 99% of bonds either remain in that decile or move to the nearest trade activity decile. Of bonds in deciles 6-10, 74% to 76% continue in the more actively traded group in the following year.

One naturally expects it will take longer for a dealer to dispose of a bond with few or no trades in the past 30 days than it will for a bond with greater recent trading activity, particularly given the stability of trading activity deciles. For each dealer roundtrip (as defined in the next section), we use the trading activity of the bond over the prior 30 days as a proxy for the dealer's expected holding period risk.¹⁴

¹³ We define deciles by pooling bond-month observations over our entire sample period so that our deciles also reflect time series variation in liquidity. Specifically, at times when overall market liquidity is lower, we will observe more roundtrips occurring in the less active trading deciles.

¹⁴ Trading activity deciles are defined using the prior 30 days' trades of *all* dealers in the bond and not just for the dealer that completes the roundtrip. Thus, these categories may overstate the liquidity experience for an individual dealer based on prior order flow, and are therefore conservative. We do, however, separately control for the prior trading and inventory behavior of the dealer in a given bond.

III.C. Identifying dealer roundtrips

Empirically, the executed transactions we observe are a combination of longer term trades where dealers take on principal risk and short term roundtrip trades where dealers act more as brokers. To identify dealer roundtrips, we follow a procedure that matches buys and sells of a dealer that are likely to be related. To more accurately match groups of trades, we include buy and sell transactions of any size since larger dealer buys might be broken up into smaller sell transactions.

We first identify all instances of a buy and sell trade for a given bond of the same quantity, from the same dealer, and within the same day. When a dealer has more than two matching trades within the day, we match the buy and sell closest in time. We refer to these pairs as “paired roundtrips.”¹⁵ After identifying paired roundtrips, we next identify instances where a given dealer’s buy trade is broken into more than one sale on the same day, by matching the buy trade with two or more sells with total quantity up to the amount of the buy trade. Because the anonymous dealer codes in our version of the TRACE data enable us to identify trades of the same dealer, we do not need to limit our analysis to near simultaneous trades; requiring trades matched within the day to occur within a smaller time interval has an insignificant impact on the descriptive or regression results in this paper. Lastly, we use all remaining transactions as follows. For each remaining dealer buy, we follow subsequent transactions in time order to locate sells of the same dealer and same bond, cumulating up to the quantity of the initial buy. Sells must occur within 60 days of the initial dealer buy and can be of any size.¹⁶

Thus, a dealer roundtrip consists either of a pair of trades that form a paired roundtrip, a group of trades that form other roundtrips completed in the same day, or other roundtrips completed in more than one

¹⁵ Our methodology for identifying this type of roundtrip is close in spirit to that of “paired trades” in Zitzewitz (2010) and Sirri (2014) and “riskless principal trades” in Ederington, Guan, and Yadav (2014), Harris (2015), and Schultz (2017), but differs in several important respects. These papers match a buy and a sell trade of a dealer of the same quantity, which are simultaneous or close in time, and which most often utilize the interdealer market for one of the trades. Further, we examine roundtrips of a given dealer that involve more than two trades as well as those that occur over longer time periods.

¹⁶ In matching trades within the day, we allow sell trades to precede the buy transaction such that trades matched are closest in time. If a buy trade is matched with a portion of a sell trade of greater quantity, the unused quantity of the sell trade can be used as part of a roundtrip occurring over more than one day. Results are insensitive to our treatment of these portions of sell trades, as the vast majority of same day round trips have a buy quantity exactly matching the total volume of sell trades.

day. The important distinction for the third group is that when roundtrips are not completed within the day, the dealer has taken the bonds overnight into inventory. Our methodology produces a dataset of 27,380,670 roundtrips for 55,988 TRACE bonds. Appendix Table A3 shows that the dealer roundtrips we identify cover approximately 93%, either by volume or trade count, of all trades reported to TRACE for these bonds.

Table 3 summarizes characteristics of all dealer roundtrips where the initial buy trade is for at least 100 bonds, which we use as the cutoff for institutional versus retail size trades.¹⁷ We include only observations of spreads for the 8,576,027 roundtrips that do not include any trades reported to TRACE as agency trades. Panel A shows that a large proportion of trading occurs in the form of paired roundtrips, which account for 41.1% of all dealer roundtrips. Another 17.1% of roundtrips are completed in the same day but not perfectly paired, with a single dealer's buy offset by a single sell trade of the same quantity, showing the significance of dealers' ability to offset inventory beyond the paired roundtrips. The remaining 41.8% of roundtrips are completed in more than one day, indicating cases where dealers take on overnight inventory risk.

We define the dealer's holding period as the weighted average number of days between the initial dealer buy and the subsequent dealer sells. When roundtrips are not completed within the same day, holding periods can be substantially longer than just overnight, with a median of 21 days for such roundtrips. When roundtrips consist of more than two trades, the weighted average holding period measure does not describe the total time until the dealer has no remaining inventory from the initial purchase that started the roundtrip. We therefore report the number of days until the last sell transaction within the roundtrip (length of holding period), which has a median of 28 days for all roundtrips completed in more than one day.¹⁸ We also report the time until the last sell trade in the roundtrip in seconds, which has a mean of 1,515 seconds (about 25 minutes) and median of 99 seconds for the paired roundtrips, showing that these trades typically occur very close in time. The mean number of sells for non-paired roundtrips completed in one day is over two,

¹⁷ We expect larger trades to be more informative in understanding dealer inventory behavior, but report characteristics for 18,973,264 smaller roundtrips in Appendix Table A4.

¹⁸ Friewald and Nagler (2016) find that dealers' holdings of TRACE bonds have a half life of 5 to 6 weeks, consistent with our findings that some bonds are held in inventory for significant lengths of time. Anand, Jotikasthira, and Venkataraman (2017) report dealer aggregate inventory cycles (starting and ending at zero inventory) last 79 days.

suggesting that only focusing on paired roundtrips with one exact offsetting trade misses about 30% of the same-day roundtrips where dealers offset their purchase with more than one counterparty. For roundtrips completed over more than one day, on average 3.27 sell trades are matched to each dealer buy, and average trade sizes are relatively large.

An important consideration is whether to interpret paired roundtrips as pre-arranged by the dealer. Characterizing trades as “prearranged” may be more appropriate for bonds that rarely trade than for more active bonds, for which dealers can offset trades through natural order flow. As with most datasets, we do not observe when a customer has initially contacted a dealer or the dealer’s search activities either before or after a trade. When we observe that a bond rarely or never traded in the recent past is quickly matched to another trade, we assume that search has occurred before executing the trade. When a bond is already held in a dealer’s inventory prior to the day on which a roundtrip starts, we assume the bond has lower search and/or inventory costs for that dealer.

IV. Dealer Roundtrip Spreads and Holding Periods

In this section, we provide initial description of roundtrip transactions costs based on measures of prior trading activity. We then examine the relationship between dealer holding periods and roundtrip costs.

IV.A. Roundtrip transaction costs

We calculate roundtrip transactions costs as the weighted average of dealer sell prices minus the dealer buy price. When the initial purchase is followed by multiple customer sales, we calculate a volume weighted average selling price. Table 4, Panel A reports initial summary statistics for roundtrip transactions costs overall and by credit rating and trading activity deciles. Overall roundtrip spreads have a median of 15 basis points (\$1.50 for a \$1000 par value bond). Spreads are clearly increasing as credit rating declines, and

their magnitude within rating categories is comparable to that reported in a number of papers using various methods to compute transactions costs.¹⁹

Inventory models predict that spreads will increase with the credit risk of the bond, and for less active bonds which expose the dealer to greater inventory risk if they remain in inventory longer (Grossman and Miller, 1988). For each roundtrip, we calculate trading activity over the 30 calendar day period ending the day prior to the start of the roundtrip. Table 4, Panel A demonstrates a striking result that shows the first elements of dealer behavior varying by trading activity. Controlling for credit rating, these univariate comparisons do not show that spreads monotonically decline with increases in trading activity as measured by trade count or days traded, as would be predicted by standard inventory models. Using trade count deciles for bonds rated A and above, the median spread for the least active decile (decile 1, 0-16 trades in the past 30 days) is 0.10, which is significantly smaller than the median of 0.125 for the most active bonds in deciles 6-10. In fact, the less active deciles 1 through 5 each have a lower median spread than the more active half of the sample. Interestingly, spreads appear quite constant (though medians are statistically different) across deciles for the lowest rated bonds.

Panel B of Table 4 reports spreads based on the type of roundtrip (paired, other same day, or completed in more than one day) as well as the type of counterparty (dealer versus a customer). The median spread on roundtrips that take longer than a day (0.29) are about three times as large as those for paired roundtrips (0.10). This is consistent with the description in Harris (2015) and Zitzewitz (2010) that paired trades have smaller spreads than non-paired trades, though spreads for our paired roundtrips are approximately half the “mean RPT markup” in Harris (2015). Table 4, Panel B further shows that over half (4,604,268, or 53.7%) of roundtrips originate with a dealer buying from another dealer, demonstrating an active interdealer market. However, for 55.6% of the paired roundtrips and 61.4% of the other same-day roundtrips, the dealer buys from and sells only to a customer. Thus, the interdealer network is not used for a

¹⁹ For example, Choi and Huh (2017) report roundtrip spreads from various estimation methods ranging from 17 to 25 basis points for investment grade bonds, and 25 basis points for high yield. Excluding simultaneously matched trades, Bessembinder et al (2017) report somewhat higher transactions costs from a regression model ranging from 0.25 pre-crisis to 0.47 during the crisis for trade sizes of \$100K.

majority of roundtrips when the dealer buys from a customer, further implying that there are not longer intermediation chains; this is in strong contrast to both Harris (2015) and Zitzewitz (2010) who find that almost all paired trades involve the interdealer market. Dealers are more likely to rely on the interdealer market for the sale when they have purchased bonds from another dealer.

In over 72% of roundtrips overall, the dealer either buys from a customer or sells of at least some portion of those bonds to customers. Models of network formation suggest that finding customers is more valuable than being a pass-through in the interdealer market. While we expect to observe higher spreads when the roundtrip includes sales to a customer, the relationship between spreads and an initial purchase from a customer versus a dealer customer is less clear. When a dealer purchases from another dealer, unless a counterparty has been prearranged, the purchasing dealer assumes the inventory risk, for which it should be compensated with a larger spread. However, if the dealer buys from a customer, it does not share rents from this portion of the chain of intermediation with another dealer.

Consistent with these predictions, spreads for paired and other same day roundtrips are very low (6 basis points) when a dealer both buys from and sells to another dealer. Similar to the results in Sirri (2014) for municipal bonds, we find that spreads are highest when the dealer buys from another dealer and sells to a customer, suggesting the value of locating a customer or that dealers demand better prices than customers. Further, when a dealer buys from another dealer and takes the bond into overnight inventory, we observe substantially higher spreads (0.35, versus 0.08 for paired roundtrips and 0.13 for other same day roundtrips), particularly when the dealer subsequently sells those bonds to another customer.²⁰

IV.B. Dealer holding periods

Our description of spreads raises an intriguing question – how does dealer behavior change with bonds' trading activity, and can this behavior explain the observed relationship between roundtrip spreads and trading activity? Given the relative illiquidity of many bonds, absent an active search for a counterparty,

²⁰ The regression results in Harris (2015) suggest that customer-dealer-customer trades have a spread about one-third of the customer-dealer-dealer or dealer-dealer-customer trades; the results here suggest otherwise.

dealers would likely have to hold bonds in inventory for a significant time until the arrival of a counterparty. Correspondingly, customers facing higher costs of immediacy may hold the bonds until a counterparty is located. As noted in the previous sections, observed roundtrip costs reflect a combination of roundtrips completed within a day, which includes cases where the dealer undergoes costly search to find offsetting trades, and roundtrips which take longer than a day to complete, for which the dealer takes on inventory risk.

In this section, we explicitly examine whether dealers appear to adjust holding periods to mitigate inventory risk, providing less immediacy their counterparties. As shown in Table 3, holding periods can be considerably longer for roundtrips not completed the same day. To investigate, Table 5 reports the percentages of completed roundtrips of each type by rating groups and trading activity deciles. Roundtrips for lower rated bonds are more often completed in the same day. For both trading activity measures, less actively traded bonds have the highest frequency of paired roundtrips, and correspondingly the lowest frequency of roundtrips that are not completed within the same day and thus held in inventory at least overnight. Higher trading activity should imply a higher likelihood of finding an offsetting trade in the same day, not less. This description therefore suggests that dealers actively seek or create offsetting trades on the same day for the least active bonds, providing further support for the hypotheses in Section II.B.²¹

To better highlight these relationships for active versus inactive bonds and higher versus lower credit quality bonds, Panel A of Table 6 repeats this description for the subsamples of bonds rated A and above versus those rated B and below. The notable contrasts are highlighted in the table. In each of the trading activity deciles, bonds rated B and below have more paired roundtrips than bonds rated A and above. Comparing higher rated, more active bonds (deciles 6-10) to lower rated bonds with less trading activity (decile 1), we consistently observe that lower rated less active bonds have a higher incidence of paired roundtrips and lower incidence of roundtrips completed in more than one day. Using trade count deciles, for example, 43% of roundtrips for higher rated more active bonds are completed over more than one day, with

²¹ The incidence of other same day roundtrips is increasing in trading activity. Particularly for more actively traded bonds, this group is most likely to include roundtrips that may not have been prearranged but for which the dealer is still able to offset within the same day. See discussion in Section V with respect to robustness of our results to classification of these roundtrips.

an average holding period of 21.9 days, while lower rated least active bonds have only 29.7% such roundtrips (with a holding period of 24.9 days). Within rating categories, the incidence of paired roundtrips is substantially higher for the least active bonds. These results strongly suggest that dealers actively search for counterparties for riskier and less active bonds which would otherwise expose the dealer to larger inventory risk. Although this is not consistent with standard microstructure models of dealers waiting for flow, it is consistent with the hypotheses in Section II.B based on active search by dealers, particularly for inactive or risky bonds.

As noted in Section II.A, a number of papers that model interdealer networks such as Neklyudov (2014), Chang and Zhang (2016), and Hollifield, Neklyudov, and Spatt (2017) suggest that individual dealers also differ in their ability to locate counterparties (search costs), their ability to manage inventory, and therefore the cost of immediacy offered to customers. Several possible variables can be used as empirical proxies for a dealer's size and position in the dealer network.²² Table 6, Panel B compares roundtrip completion of top volume to other dealers, and shows that top dealers have a lower frequency of paired roundtrips; results are similar (not reported for brevity) using *top ten dealer: trade count* or top 1% dealers. As Di Maggio, Kermani, and Song (2017) and others note, market trading is extremely concentrated with these top dealers. While they may pre-arrange trades, top dealers are also most likely to already have bonds in inventory, or can take bonds into inventory and quickly offset the trade because they are active market makers. From Panel B of Table 6 we observe these top dealers still match trades more often for the less active bonds; however, compared to other non-top dealers they have a lower frequency of paired roundtrips and more roundtrips completed in more than one day. Table 6, Panel B also shows that the frequency of roundtrips where top dealers take bonds into inventory is more similar across trading activity deciles.

²² Network measures using interdealer trades are estimated by Di Maggio, Kermani, and Song (2017) for corporate bonds, Hollifield, Neklyudov, and Spatt (2017) for structured products, and Li and Schürhoff (2015) for municipal bonds. These and other papers demonstrate the extreme concentration of trading among a small number of dealers, consistent with a core-periphery network structure. Our regressions in the following sections include a continuous measure of network centrality (eigenvector centrality), as well as individual dealer's prior trading and inventory behavior for a given bond.

Our results are also consistent with Chang and Zhang (2016) who suggest that top dealers have lower risk sharing needs, therefore take bonds into inventory and are less sensitive to prior liquidity considerations. Panel B of Table 6 shows that the holding period for roundtrips of top dealers is also longer than for others, particularly for less active bonds (for example, 21.3 days for top dealers versus eight days for other dealers for the lowest trade count decile). These results point to large differences in behavior for the top dealers that account for a substantial proportion of trading in this market.

V. Dealer Endogenous Search

Our findings thus far are consistent with the conjecture that dealers actively mitigate their risk by managing their holding periods. Searching for a counterparty or pre-arranging a counterparty prior to trade is costly, but reduces the risk of taking the bond into inventory. Different dealers face different inventory risk based on the natural flow they receive from other dealers.

The problem facing the dealer is similar in spirit to the electronic versus voice choice for trading corporate bonds in Hendershott and Madhavan (2015).²³ We model the choice of holding period and roundtrip spreads using endogenous switching regressions. The first stage selection model (reported in this section) is estimated as a probit, where the dependent variable is an indicator for roundtrips completed within the same day. The second stage regressions (reported in Section VI) examine the resulting spreads based on whether the roundtrip is completed within a day versus the bonds are held at least overnight in inventory.

The first stage model includes variables we expect to be related to the decision to undergo costly search for a counterparty, which would result in a roundtrip being completed as a paired trade or otherwise within the same day. We estimate the following probit model, where the dependent variable equals one for paired roundtrips or other roundtrips completed within the same day:

²³ Hendershott and Madhavan (2015) study the use of an electronic request-for-quote system developed by Market Axess, on which investors can query multiple dealers electronically and select the best quote. They show that the use of this system is concentrated in bonds most likely to be more liquid (i.e. higher quality, younger, shorter term, and larger issues). A recent survey by Greenwich Associates (2015) estimates that only 4% of corporate bond volume is traded electronically for the “less liquid corners, including high yield.”

$\Pr(\text{same day} = 1)$

$$\begin{aligned}
 &= \alpha + \beta_1 * d_1 + \beta_2 * d_2 + \beta_3 * d_3 + \beta_4 * d_4 + \beta_5 * d_5 + \beta_{6-10} \\
 &\quad * (\text{bond specific characteristics}) + \beta_{11-15} * (\text{dealer characteristics}) + \beta_{16-21} \\
 &\quad * (\text{trade and counterparty characteristics}) + \beta_{22-23} * (\text{market conditions}) + \varepsilon
 \end{aligned}$$

(Eqn 1)

Appendix Table A5, Column 1 summarizes the empirical proxies used in this model as they relate to our hypotheses in Section II, as well as the predicted effect on the probability of completing a roundtrip in the same day. We estimate our model separately for rating groups; the lowest credit quality bonds have greater price risk and therefore inventory risk, are more often Rule 144A bonds, and often have lower trading frequency relative to investment grade bonds (Goldstein and Hotchkiss, 2009). Thus we expect dealers have a greater likelihood to complete roundtrips within the same day for lower rated bonds, particularly those which trade infrequently.

Of primary interest are the trading activity decile dummies, d_1 through d_5 : decile 1 indicates the least active decile and the omitted group is the top five deciles (6 through 10) of more actively traded bonds.²⁴ Using dummies for trading activity deciles also allows for possible nonlinearities. In addition, bond characteristics related to higher expected inventory or search costs (greater time to maturity, Rule 144A bonds, and older bonds) should increase the likelihood a roundtrip is completed the same day. The effect of issue size is unclear, as larger issues may be easier to locate, but dealers may be more willing to hold them in inventory. Search costs, and therefore the expected inventory holding period, are also expected to be lower

²⁴ Although it cannot be used to estimate selection effects for the second stage model, we also run a multinomial logit in order to further understand differences between paired roundtrips and other roundtrips completed the same day. The dependent variable indicates each of the three types of roundtrips, where the holdout group is paired roundtrips. As expected, paired roundtrips have a higher likelihood relative to other same day roundtrips for lower trading activity deciles. Coefficients for roundtrips completed over more than one day (versus paired roundtrips) are similar to those in Table 7. The multinomial logit results suggest that combining paired roundtrips with other same day roundtrips yields a conservative description of the differences in dealer behavior across trading activity deciles.

We also re-estimate our endogenous switching regression models, either dropping other same day roundtrips from our analysis or combining these roundtrips with roundtrips completed in more than one day. Results using this alternative classification are similar to those reported here. The specifications we report, however, better represent the comparison of roundtrips based on whether the dealer takes on overnight inventory.

when more dealers trade the bond; we include the natural log of the number of dealers trading in the bond over our sample period. We further expect that dealers are more likely to take bonds into inventory when their ability to hedge is better; hedging costs are expected to be lower for bonds where the issuer's stock is listed, and for bonds with traded CDSs (as indicated by CDS quotes available from Markit).

Based on the models of dealer search and networks described in Section II.A, top ten dealers (by total trading volume or by total trade count) and better connected dealers (measured by eigenvector centrality) face lower search costs. These dealers may more quickly locate a counterparty within the same day, even if the trade has not been prearranged. At the same time, these dealers likely benefit from lower funding costs, and may be more likely than other dealers to take bonds into inventory. Still, even larger and better connected dealers face relatively higher costs for riskier (lower rated) and less actively traded bonds. We include indicator variables for whether the dealer is a top ten dealer by either volume or trade count as well as a measure of the dealer's eigenvector centrality. Results throughout this paper are insensitive to other measures of the dealer's position in the network (degree, in-degree, out-degree) which are each highly correlated with eigenvector centrality (Hochberg, Ljungqvist, and Lu, 2007).

The selection equation includes two additional dealer-specific variables that we expect to be associated with a lower probability of completing the roundtrip within the same day. The first, *dealer takes inventory (this bond)*, using all trades of a given dealer in a specific bond starting in the week prior to the start of the roundtrip, and is calculated as the number of days the dealer has taken the bond into overnight inventory divided by the total number of prior days the dealer has traded the bond. The second, *dealer traded in prior year (this bond)*, is calculated as the number of days the dealer has traded this bond in the prior year (days -6 to -371 relative to the start of the roundtrip), and is an indication of the dealer's natural order flow in the bond; a low value suggests the dealer likely has higher search costs.

We control for possible non-linear effects of size of the dealer's buy trade by including three size dummies, which correspond to the TRACE dissemination volume cap breakpoints (the holdout group

includes bond purchases from 100 to 499 bonds).²⁵ Larger initial purchases by dealers may take longer to unwind, so that dealer inventory risk should increase with the size of the buy trade. The outside opportunities of the investor cannot be easily inferred from the TRACE data as customer identities are not provided; however, our analysis includes only roundtrips starting with a dealer buy of at least 100 bonds which likely originate with a purchase from an institutional investor. Lastly, we include an indicator for roundtrips where the dealer purchases bonds from a customer rather than another dealer (*buy from customer*), as previously described in Section IV.B.

The results of the selection equation are reported in Table 7. Dealers are generally more likely to complete transactions within a day for the *least* active bonds, consistent with active search by dealers and with our earlier hypotheses. The coefficient for decile 1 (least active bonds) is economically much larger relative to more active bond deciles. Calculating marginal effects from the reported coefficients and using days traded as the measure of trading activity, roundtrips for A rated bonds (B and below rated bonds) in decile 1 have a 12% (11%) higher likelihood of being completed the same day. Also using deciles based on the number of days traded, for all rating groups the likelihood of completion the same day declines monotonically through decile 3 as trading activity increases. Since the likelihood of a counterparty arriving should decrease for less active bonds making a dealer roundtrip within a day less likely, these results strongly suggest that dealers undertake costly search to find counterparties in order to mitigate risk.

Other reported variables affect selection as well. As credit quality falls, roundtrips starting from a larger buy trade are more likely completed the same day. Roundtrips of Rule 144A bonds, older bonds of lower credit quality companies, and larger issues also have a higher probability of completion the same day. The ability to hedge bonds appears important for lower credit quality bonds (*listed stock*; *CDS trades*). The probit results also show that top dealers based on volume more often take bonds into inventory. Interestingly, the coefficient for the variable measuring top dealers by their number of trades has the opposite sign, but only for investment grade bonds; for non-investment grade bonds there is no statistical effect. This likely

²⁵ TRACE does not display volumes for trades over \$5 million for investment grade and \$1 million for non-investment grade bonds during our sample period. Our dataset provides uncapped volumes.

reflects that these are more peripheral dealers with a greater proportion of smaller sized retail trades which are more likely to transact in investment grade bonds, or that search costs and/or risk sharing issues are greater when transaction sizes are larger. Consistent with dealer network models, more central dealers (*eigenvector centrality*) are less likely to complete roundtrips the same day, but the effect is less strong for lower rated bonds which pose greater inventory risk even for better connected dealers.

We also observe an interesting effect of the size of the initial dealer buy for higher quality bonds; as trade size increases, dealers are more likely to take the bond into inventory, as would be expected if such trades take longer to unwind. But, the opposite is true for roundtrips starting from larger purchases of non-investment grade bonds which are more likely completed the same day, again strongly consistent with dealers mitigating inventory risk.

The probit models also show that the probability that a dealer completes a roundtrip within a day increases if the bond is initially purchased from another dealer (negative coefficient on the indicator variable *buy from customer*), although this effect falls for lower rated bonds. Dealers are less likely to purchase a bond from another dealer unless there is a high likelihood of finding another customer, so within day roundtrips are more likely when the dealer buys from another dealer. Lastly, during the crisis period, roundtrips are less likely to be completed the same day (relative to the pre-crisis period). Post-crisis, this effect is reversed. The increase in roundtrips completed the same day during the crisis is consistent with recent studies showing a reduction in aggregate bond inventory during the crisis (Choi and Shachar, 2016; Dick-Nielsen and Rossi, 2017). Interestingly, the effects are greater for higher rated bonds; calculating marginal effects using either trading activity measure, the probability of completion the same day for high rated bonds rises by 11% for the crisis period (we further examine the relationship between dealers' behavior and prior trading activity during the crisis in Section VII.B below.) Overall, the first stage probit results are consistent with the hypotheses in Section II.B that dealers are more likely to pre-arrange trades when their expected inventory costs are high.

VI. Dealer Roundtrip Spreads Controlling for Selection

The second stage of our endogenous switching model consists of two equations for spreads of roundtrips completed the same day or over longer holding periods. For each rating category, we estimate the weighted average dealer spread on roundtrip i as:

$$\begin{aligned} Spread_{(same\ day=1,i)} &= \alpha + \beta_1 * d_1 + \beta_2 * d_2 + \beta_3 * d_3 + \beta_4 * d_4 + \beta_5 * d_5 + \beta_5 * d_5 + \beta_6 * X_i \\ &\quad * InvMillsRatio_{1i} + \mu_1 \end{aligned} \quad (eqn\ 2)$$

$$\begin{aligned} Spread_{(same\ day=0,i)} &= \alpha + \beta_1 * d_1 + \beta_2 * d_2 + \beta_3 * d_3 + \beta_4 * d_4 + \beta_5 * d_5 + \beta_5 * d_5 + \beta_6 * X_i \\ &\quad * InvMillsRatio_{0i} + \mu_2 \end{aligned} \quad (eqn\ 3)$$

where same day = 1 if $I(\text{same day}^* > 0)$, as estimated from the first stage probit (equation 2), and $\text{corr}(\epsilon, \mu) = \rho$ (same day^{*} is the unobservable latent variable; see Maddala, 1983). Each equation includes the inverse Mills ratio from the stage 1 probit.²⁶

The vector of explanatory variables X_i includes bond, dealer, counterparty, and trade characteristics related to expected inventory and search costs, as summarized in Column 2 of Appendix Table A5. We expect roundtrip spreads to be lower for larger issues and issues traded by more dealers. Whether larger and better connected dealers trade at lower spreads is an empirical question and depends on the type of asset traded, characteristics of the customers, and market conditions.²⁷

Dealer sells may include smaller trades of less sophisticated or connected retail traders. The second stage models therefore include a variable indicating the proportion of sell transactions in the roundtrip that

²⁶ To reduce the effect of outliers, we eliminate roundtrips where the bond price is greater than \$150 or less than \$10, and outliers of calculated spreads at the 1% level. Our regression results are unchanged when we control for broader market movements on days while the bond is in the dealer's inventory, or when we restrict the group of roundtrips not completed the same day to those completed in shorter holding periods.

²⁷ Hollifield, Neklyudov, and Spatt (2017) find a centrality discount in the market for structured products. Li and Schürhoff (2015) find a centrality premium for municipal bonds, as do Di Maggio, Kermani, and Song (2017) for corporate bonds.

are less than 100 bonds (*ratio small/total # sells*), which is zero for paired roundtrips and often zero for other roundtrips completed within the day. We also expect to observe higher spreads when the roundtrip includes sales to a customer (*% sale volume to customers*); models of network formation such as Condorelli, Galeotti, and Renou (2017) suggest that finding customers is more valuable than being a pass-through in the interdealer market. The relationship between spreads and an initial purchase from a customer versus a dealer customer (*buy from customer*) is less clear. When a dealer purchases from another dealer, unless a counterparty has been prearranged, the purchasing dealer assumes the inventory risk, for which it should be compensated with a larger spread. However, if the dealer buys from a customer, it does not share rents from this portion of the chain of intermediation with another dealer.

Results are reported in Table 8. The importance of selection effects of completing a same-day roundtrip vs. taking a bond into inventory from the first state probit in Table 7 is demonstrated by the coefficient for the inverse Mills ratio, which is statistically significant for all credit ratings and for both regressions in Panel A (trade count) and Panel B (days traded) of Table 8. Within the group of roundtrips that are prearranged, as well as within roundtrips where the dealer takes the bond into inventory, we expect to see higher spreads for the lower trading activity deciles once we control for search effects through the selection criteria. As in the first stage probit analysis in Table 7, roundtrips observed to be completed within the same day include both those for which the dealer more readily locates a counterparty, and those for which the dealer has pre-arranged the offsetting trades; the former is more likely to occur for higher quality bonds and more actively traded bonds.

Panel A of Table 8 also shows that while spreads are higher for less active bonds based on the Decile 1 indicator for bonds rated A and above for roundtrips completed in the same day and those taken into inventory, spreads on the least active bonds (Decile 1) continue to be statistically insignificant different from the top 50% most active (Deciles 6-10) for same day roundtrips for all but the highest rated bonds (i.e., bonds rated BBB and lower). For roundtrips taken into inventory, spreads are only slightly higher for Decile 1 for bonds rated BBB and lower. With the exception of bonds rated A and above, generally bonds in the less active deciles 2 to 5 continue to have the same or lower spreads than bonds in the top 50% of activity

(Deciles 6-10). The results in Panel B (days traded) are generally similar: besides the least traded deciles 1 and 2, roundtrip spreads are generally insignificantly different than the spreads of the top 50% (deciles 6-10) by days traded for all but the highest rated bonds. These results are in contrast to the OLS results in Harris (2015), which suggested that spreads decreased as liquidity (total trades or days traded) increased, and instead support our main results in Table 4 regarding spreads and prior trading activity.

The additional control variables generally have signs as predicted. Notably, we observe that spreads are lower for more central dealers based on either volume or eigenvector centrality. These results, suggesting a centrality discount, differ from those found by Di Maggio, Kermani, and Song (2017) for corporate bonds. Measuring top dealers based on trade count picks up a number of lower volume dealers with a large number of small trades; the coefficient for this variable is positively related to spreads, and may explain the difference in our results.

Spreads are significantly greater for roundtrips starting from a larger dealer purchase. We also observe higher spreads when dealers sell to customers rather than into the interdealer market, and when some sells in the roundtrip are for less than 100 bonds (small sells). Overall, for all but the highest rated bonds, both Panels A and B suggest that spreads do not monotonically increase as liquidity decreases, even after controlling for selection effects. For the least liquid bonds in Decile 1, inventory effects likely swamp search costs and spreads become higher.

The second stage estimates can be used to compute the counterfactuals - what is the expected spread on a roundtrip actually completed the same day if it were instead brought into inventory and held longer? For roundtrips completed the same day, this is calculated as the observed spread minus the spread predicted using the estimated coefficients from equation 3 (roundtrips completed in more than one day). We report the mean difference as “observed minus predicted from > 1 day” in Table 8. For both trading activity measures, this shows that spreads would have been larger (predicted is greater than observed) if the dealer had not found a same-day counterparty and instead had taken the bond into inventory. For example, for trade count (Panel A), spreads are predicted to be 13 basis points greater had the roundtrip not been completed in the same day for bonds rated A and above, and 9.9 basis points higher for bonds rated B and below (all reported means in

the table are significantly different from zero at the one percent level). Similarly, “observed – predicted from same day” shows that for bonds rated BBB and below, roundtrips that took longer than a day are predicted to have lower spreads had the dealer found a counterparty that day. For bonds rated A or above, the predicted spreads would be slightly higher had the roundtrip been completed the same day (0.9 or less basis points), possibly because it is easier for dealers to find a counterparty for higher rated bonds.

The regression coefficients in these specifications compare of dealer behavior for bonds in each of the lower deciles of trading activity to those in the top 50% of trading activity (deciles 6-10). Table 4, however, shows that bonds with trading activity in the top 50% often have trade counts hundreds of times greater those in the lowest decile, and can trade most days in the prior month versus trading zero or few days as with the lowest deciles. We therefore re-estimate our model but include only the decile 1 trading activity dummy (deciles 2-10 are the holdout group). We also re-estimate our model but comparing roundtrips in decile 1 to all roundtrips in deciles 2 through 5 by excluding roundtrips in deciles 6 to 10 from the sample. Results are shown in Appendix Table A6, reporting only coefficients for the decile 1 indicator based on trade count and for the inverse Mills ratio (interpretation of coefficients for other included variables remains similar to those in Tables 7 and 8). Similar analysis using deciles by days traded produces nearly identical results. Together with our base specifications, these results quite clearly point to differences in dealer behavior for the least active decile of bonds, even in comparison to other relatively inactive issues.

Overall, our results are consistent with a dealer search model in which customers face a tradeoff between immediacy and execution cost. Absent search, dealers would require a higher spread as compensation for bonds more likely to remain longer in inventory. When dealers search and successfully find a counterparty, we observe a lower roundtrip spread. However, dealers may face high search costs or even fail to locate a counterparty, particularly for bonds rated BBB and below. Our results show that as liquidity and credit quality decrease, search intensifies (Tables 5 and 6); as we show, the empirically observed roundtrip spreads will not be consistently related to liquidity if search versus expected inventory costs are balanced in equilibrium.

VII. Market stress and non-top dealers

In this section, we provide an analysis of dealer behavior in periods of market stress, defined as the financial crisis period and days of large changes in the VIX index. We also examine how behavior differs when non-top dealers trade relatively inactive bonds. Lastly, we consider the impact of changes in transparency regimes under TRACE.

VII.A. Results during periods of market stress

We first consider whether dealer behavior differs in periods of market stress by examining roundtrips during the financial crisis period, which we define as the second quarter of 2007 through the second quarter of 2009. The first stage results in Table 7 suggest that there is a significantly higher propensity to complete roundtrips the same day during the crisis period, suggesting that dealers engaged in active search to reduce their risk during this period. We examine how this change impacts less actively traded bonds by including interactions of the crisis dummy with each trading activity decile. For brevity, we report only results using trade count deciles of trading activity; results using days traded are qualitatively similar.

Several earlier papers show that spreads are higher during the crisis (Bao, Pan, and Wang, 2011; Dick-Nielsen, Feldhütter, and Lando, 2012; Friewald, Jankowitsch, and Subrahmanyam, 2012). However, our second stage results, reported in Panel A of Table 9, show this is the case only when bonds are taken into inventory. Further, higher spreads on bonds taken into inventory are only observed for the lowest trading activity deciles with the exception of the bonds rated A & above, where the interaction of the crisis dummy and trading activity deciles is large and significant for each of deciles 1 to 5. Thus, for higher credit quality bonds during the crisis period, dealers more often complete roundtrips without taking on inventory risk, but when dealers do take on inventory risk, there is a substantially stronger relationship between trading activity and spreads.

To examine whether the impact of market stress on dealer behavior is observed more broadly during our sample period, we provide a similar analysis interacting the trading activity deciles with an indicator for days of large changes in the VIX. As in Anderson and Stulz (2017), we define days of large VIX changes as

those where the increase is in the top 95th percentile of changes over our sample period. Results are again extremely similar for both trading activity measures, and are reported for the second stage model using deciles by trade count in Table 9 (Panel B). Similar to our results for the crisis period, we find that the interactions are important for higher rated bonds for roundtrips completed in more than one day, in this case for bonds BBB and above.

These results provide interesting insights into changes in dealer behavior in periods where liquidity has been shown to steeply decline. The effects for higher rated bonds are consistent with other work examining the crisis period that argues that higher rated securities become information sensitive at such times, so that market behavior becomes more similar to that for lower rated securities (Dang, Gorton, and Holstrom, 2013; Benmelech and Bergman, 2017). In a period where dealers are less willing or able to provide capital, even higher rated bonds that might otherwise be taken into inventory may be traded only when dealers successfully find a counterparty to quickly offset their risk. Further, even when observed roundtrips occur within the same day, dealers may buy bonds from customers at a greater discount to provide this service.²⁸

VII.B. Analysis of non-top dealers

Roundtrips will be completed the same day when a dealer first locates an offsetting counterparty, but they may also reflect trades where the dealer maintained inventory in the bond but was able to quickly offset the trade. Further, given the concentration of trading with the largest dealers in the market, our prior results suggest that behavior differs for less well connected dealers facing greater search costs and likely higher inventory risk. Therefore, in Table 10 we report coefficients for the decile indicators in the two stage model, using only roundtrips of dealers which are not *top ten: volume* dealers. Using trade count deciles, spreads on roundtrips for the Decile 1 observations are significantly greater only for bonds rated A & above, and for BBB roundtrips completed the same day. Using deciles based on days traded, only the indicator for Decile 1

²⁸ Bao, O'Hara, and Zhou (2018) and Bessembinder, Jacobsen, Maxwell, and Venkataraman (2017) also discuss dealer capital commitment during the financial crisis and subsequent periods.

is significant across rating groups, and all decile indicators are positive and significant for the highest rated bonds. We note that the coefficient for eigenvector centrality remains negative and significant even when removing the most connected dealers which account for a large share of overall trading. Overall, we find that dealers' endogenous switching to complete roundtrips in the same day and the resulting impact on observed spreads is of large importance when non-top dealers trade riskier bonds and can result in an equilibrium where observed spreads appear invariant to expected liquidity.

VII.C. Changes in transparency under TRACE

An important aspect of the introduction of TRACE was the staggered phase-in of post-trade dissemination of transaction information, providing researchers an opportunity to study the impact of transparency on various measures of liquidity. While initial studies shows benefits of declining transactions costs, dealers have still questioned the impact of transparency on dealers' capital commitment, particularly in for less active bonds and in stressful markets (Bessembinder and Maxwell, 2008). Our study is useful in this context to the extent we can ask whether introduction of transparency impacts dealer behavior, especially for less actively traded bonds.

The difficulty in implementing such tests is that the phase-in of dissemination is based on specific bond and trade characteristics (see timelines in Goldstein and Hotchkiss, 2009, and Asquith, Covert, and Pathak, 2013). To the extent our tests control for each of the specific bond characteristics used to determine whether trade information is disseminated, we already control for the effects of transparency.²⁹ Still, in Appendix Table A7, we provide limited evidence that for bonds rated B or below, dealers are more likely to take bonds into inventory and roundtrips spreads are lower when the least active bonds become disseminated (triple interaction of disseminated bond, post-dissemination, and decile 1). For all other rating categories, no significant effects are observed for the interacted variable.

²⁹ Over the multiple phase in periods for TRACE these include issue size, credit rating, the frequency of prior trading, and the Rule 144A status of the bond; see Goldstein and Hotchkiss (2009) for description of the staggered introduction of dissemination.

Given it is difficult to identify the impact of transparency independent of specific bond characteristics, we are cautious in our interpretation. Even so, these results do not support the concern that transparency itself negatively affects dealer capital commitment to less actively traded bonds. In general, dealers may be more likely to offset purchases of such bonds within the same day even in opaque regimes.

VIII. Conclusion

Many corporate bonds trade rarely if at all: one or two trades over the past 30 days across all dealers in a bond is not unusual. For an individual dealer, trading is even less frequent. However, little is known about dealer behavior in markets where many assets trade infrequently. Standard market microstructure models, which model dealers as passive order takers waiting for customers to arrive, suggest that dealers will hold less actively traded assets in inventory for a long time, subjecting the dealer to greater price risk and likely requiring a large bid-ask spread as compensation. Applied to the corporate bond market, this implies that a dealer who cannot immediately sell to another investor or dealer is exposed to inventory risk. Since search activity can mitigate this risk, we expect market makers to increase their search intensity for less actively traded and lower credit quality bonds, acting more as brokers and correspondingly providing less immediacy to customers through capital commitments.

Using a comprehensive sample of U.S. corporate bonds having trades reported to TRACE, we examine dealer behavior in a market where many securities are inactively traded. We find that dealers are significantly less likely to take less active bonds into inventory, and instead find same-day counterparties. We estimate an endogenous switching regression model to estimate spreads on dealer roundtrip trades. Our model accounts for dealers' ability to adjust their behavior when faced with a tradeoff between costly search versus costly inventory holdings. Collectively, our empirical results are consistent with a dealer search model in which customers face a tradeoff between immediacy and execution cost. Overall, we find that dealers balance search and inventory costs in such a way that dealer roundtrip costs do not increase with expected illiquidity.

Our results that dealers' behavior endogenously changes based on expected liquidity are important in interpreting a number of recent studies that seek to explain observed trading costs for corporate bonds, either cross-sectionally or over time. The response of dealers to other recent developments that potentially impact market liquidity therefore needs to be considered in evaluating overall changes in market quality. For example, Choi and Huh (2017) and Anand, Jotikasthira, and Venkataraman suggest that buy-side institutions are becoming more important liquidity providers. Our analysis provides guidance as to how expected dealer behavior and observed trading costs may respond to innovations affecting market liquidity.

Our results also speak to the debate regarding the impact of transparency on corporate bond liquidity. Some market participants suggest that comparisons of spreads or other measures of liquidity in pre- and post-dissemination regimes overlook the question of whether dealers become more reluctant to commit capital for making markets in illiquid securities in transparent markets (Bessembinder and Maxwell, 2008). However, even in an opaque market, dealers appear to search for counterparties to mitigate risk as illiquidity increases. To the extent dealers already commit little capital to less actively traded issues, concerns that capital is withdrawn from this portion of the market due to increases in transparency become less relevant.

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Appendix Table A1
Variable definitions

Bond characteristics	Rating	Numeric scale for credit rating at the time of trade, where 1=AAA, 2=AA, etc. Unrated bonds are assigned to the lowest rating quality group (11). Ratings are based on S&P, Moody's, and Fitch as available, and are consistent with original FINRA guidelines for bond dissemination under TRACE.
	Bond age	Time since bond issuance (years) at time of dealer roundtrip
	Issue size	Offering amount
	Rule 144A bond	Indicator variable for non-publicly registered Rule 144A bonds
	Time to maturity	Time remaining from the start of the dealer roundtrip to the bond maturity
	Ln(# dealers)	Log of number of dealers in the bond based on all trades reported to TRACE over the full sample period.
	Change in aggregate bond inventory (-30,0)	Change in inventory summed for all dealers in the bond in the 30 day period ending the day prior to the start of the dealer roundtrip
	Listed stock	Issuer's stock is exchange listed (NYSE/AMEX or Nasdaq)
	CDS quotes	Quotes for the issuer's CDS are available (from Markit) prior to the start of the dealer roundtrip
Trading activity variables	Trade count	Number of trades, using all TRACE reported trades in the 30 day period ending the day prior to the start of the dealer roundtrip
	Days traded	Number of days with non-zero trading, using all TRACE reported trades in the 30 day period ending the day prior to the start of the dealer roundtrip
	Decile 1 to 5,6-10	Trading activity (trade count, days traded) in the 30 day period preceeding the dealer roundtrip is in corresponding decile using all dealer roundtrips. Decile 1 is the lowest trading activity group. Deciles 6-10 are reported as one group and include all roundtrips with trading activity above the sample median.
Dealer roundtrips	Paired roundtrips	Paired roundtrips consist of a buy and sell transaction of the same dealer and same bond, day, and quantity. When a dealer has more than two matching trades within the day we match closest in time.
	Other same day roundtrips	Using trades which are not part of a paired roundtrip, within the day we match each dealer buy transaction to sale transactions, using sells up to the volume of the starting buy. Sells can be of any size, and total volume of sells can be less than the starting buy volume.
	Roundtrips completed over more than one day	Using trades which are not part of a paired or other same day roundtrip, we match each dealer buy transaction to subsequent sale transactions, using sells up to the volume of the starting buy. Sells must occur within 60 days of the intial dealer buy. Sells can be of any size, and total volume of sells can be less than the starting buy volume.
	Buy volume >= 5000	Indicator that the dealer buy in the roundtrip is >= 5000 bonds (\$5 million)
	1000 <= buy volume < 5000	Indicator that the dealer buy in the roundtrip is < 5,000 bonds and >= 1000 bonds
	500 <= buy volume < 1000	Indicator that the dealer buy in the roundtrip is < 1,000 bonds and >= 500 bonds
	Buy from customer	Indicator that dealer buy in the roundtrip is a purchase from a customer (versus from another dealer)
	% sale volume to customers	Percentage of volme of dealer sells in the roundtrip that are to customers (versus to other dealers)
	Ratio small/total # sells	Ratio of the number sells of less than 100 bonds to the total number of sells included in the dealer roundtrip.
Dealer characteristics	Weighted ave spread	Difference between buy price and sell prices in the dealer roundtrip. Price differences are weighted by the volume of the sell trade divided by the total volume of sell trades in the roundtrip.
	Top ten dealer: trade count	Indicator for the ten dealers with the highest trade count across all TRACE dealers, summing all of the dealer's trades in sample bonds reported over the full sample period
	Top 1% dealer: trade count	Corresponding indicator variable, indicating the highest 1% trade count across all TRACE dealers
	Top ten dealer: volume	Indicator for the ten dealers with the highest total volume across all TRACE dealers, summing all of the dealer's trades in sample bonds reported over the full sample period
	Top 1% dealer: volume	Corresponding indicator variable, indicating the highest 1% volume across all TRACE dealers
	Small trade dealer	Indicator for dealers having more than 50% of trades for less than 100 bonds, based on all of the dealer's trades in TRACE
	Dealer centrality	Eigenvector centrality, calculated by dealer and by year
	Dealer traded in prior year (bond specific)	# days dealer traded this bond in year prior to the start of the roundtrip (days -6 to -371)
	Dealer takes inventory (bond specific)	# days (prior to start of roundtrip) the dealer took this bond into inventory / # days dealer traded this bond

Appendix Table A2

Stability of trading activity deciles

Table reports the frequency distribution of bonds moving from deciles of trading activity in the prior period (shown in columns) to deciles in the current period (shown in rows). Rows sum to 100 percent. The "no trade" group includes observations with no trading in the current or prior period. Percentages include all sample bond-year or bond-month observations excluding periods within six months of issuance, maturity, or redemption. Deciles are determined by ranking the median trading activity of the bond in a given period.

		Median trading activity decile by year							Median trading activity decile by month						
		Trade count							Days traded						
		Decile in the prior period							Decile in the prior period						
		No trading	1	2	3	4	5	6-10	No trading	1	2	3	4	5	6-10
Decile in the current period:	No trades	71.1	25.0	2.4	0.9	0.4	0.2	0.1	82.7	16.6	0.6	0.1	0.0	0.0	0.0
	1	7.3	85.4	6.3	0.7	0.2	0.1	0.1	21.8	65.0	11.0	1.8	0.3	0.1	0.1
	2	2.1	32.1	46.2	16.2	2.6	0.6	0.2	3.4	38.0	37.7	16.4	3.5	0.7	0.3
	3	1.6	9.3	26.6	39.5	17.5	4.0	1.4	1.5	11.8	29.1	35.6	16.6	4.1	1.3
	4	1.4	4.2	10.6	23.8	33.3	19.2	7.5	1.1	3.8	10.8	27.7	33.6	17.7	5.4
	5	1.2	2.8	4.3	11.7	20.8	31.6	27.7	1.1	1.9	3.5	10.1	24.5	35.6	23.4
	6-10	1.4	1.3	2.0	3.7	6.3	12.0	73.5	1.0	0.7	1.1	2.0	4.2	11.4	79.7
Decile in the current period:	No trades	71.1	24.7	2.7	0.8	0.4	0.1	0.1	82.3	16.9	0.4	0.0	0.0	0.0	0.4
	1	7.5	85.5	6.3	0.5	0.2	0.0	0.1	22.6	64.6	11.2	1.3	0.3	0.0	0.1
	2	1.7	27.8	51.3	14.8	3.4	0.4	0.5	2.5	35.7	41.2	13.9	5.4	0.7	0.7
	3	1.3	6.4	29.5	34.4	23.0	2.7	2.7	1.2	9.9	33.9	26.9	19.7	3.7	4.7
	4	1.3	2.6	11.5	22.0	36.1	12.6	14.0	1.3	3.1	15.8	23.4	29.7	9.0	17.7
	5	1.4	1.8	5.0	9.1	24.3	17.3	41.1	3.9	1.2	6.1	13.0	26.4	11.6	37.8
	6-10	1.4	1.0	1.7	3.0	8.9	7.5	76.4	3.6	0.7	1.8	4.0	12.1	8.7	69.1

Appendix Table A3

Comparison of dealer roundtrip volume and trade count to all TRACE reported trades

The volume and number of trades that are part of a dealer roundtrip are compared to all TRACE reported trades in the same set of bonds. The sample consists of 55,988 bonds for which at least one roundtrip is identified. Statistics are also reported for the dataset deleting all dealer buys of less than 100 bonds.

All trades on TRACE

Total \$ volume of trades that are part of a dealer roundtrip	54,672,244,140
Total # trades that are part of a dealer roundtrip	59,513,429
Total \$ volume of trades that are part of a dealer roundtrip completed same day	12,968,980,709
Total # of trades that are part of a dealer roundtrip completed same day	17,942,690
Total \$ volume reported on TRACE	57,953,409,971
Total # of trades reported on TRACE	63,864,204
All dealer roundtrip trades as percentage of total TRACE volume	94.3%
All dealer roundtrip trades as percentage of total number of TRACE trades	93.2%
Same day roundtrips as percentage of total TRACE volume	22.4%
Same day roundtrips as percentage of total number of TRACE trades	28.1%

All trades on TRACE excluding buys < 100 bonds

Total \$ volume of trades that are part of a dealer roundtrip	53,807,340,724
Total # trades that are part of a dealer roundtrip	21,401,029
Total \$ volume of trades that are part of a dealer roundtrip completed same day	12,698,666,257
Total # of trades that are part of a dealer roundtrip completed same day	6,998,849
Total \$ volume reported on TRACE	57,026,177,683
Total # of trades reported on TRACE	23,002,055
All dealer roundtrip trades as percentage of total TRACE volume	94.4%
All dealer roundtrip trades as percentage of total number of TRACE trades	93.0%
Same day roundtrips as percentage of total TRACE volume	22.3%
Same day roundtrips as percentage of total number of TRACE trades	30.4%

Appendix Table A4

A. Summary description of dealer roundtrips: roundtrips starting from small buy trades

Characteristics of dealer roundtrips starting from a buy trade of less than 100 bonds. Variables are as defined in Appendix Table A1. Paired roundtrips are defined as a pair of trades where a given dealer buys and sells the same quantity of the same bond on the same day. Other roundtrips match each dealer buy of a given bond with one or more subsequent sell trades by that dealer in the same bond up to the face amount of the initial buy trade, using sell trades that occur within 60 days. Roundtrips which include any agency trades are excluded.

	All roundtrips (n=18,804,643)		Paired roundtrips (n=9,670,872; 51.4%)		Other same day roundtrips (n=2,185,768; 11.6%)		Roundtrips completed in more than one day (n=6,948,003; 36.9%)	
	<u>Mean</u>	<u>Median</u>	<u>Mean</u>	<u>Median</u>	<u>Mean</u>	<u>Median</u>	<u>Mean</u>	<u>Median</u>
Weighted ave holding period	9.0	1.0	1.0	1.0	1.0	1.0	22.7	18.7
Length of holding period	9.8	1.0	1.0	1.0	1.0	1.0	24.9	22.0
Time to last sell (seconds)	763,196	1,695	806	2	6,889	4,560	2,062,287	1,796,156
# sells in the roundtrip	1.22	1.00	1.00	1.00	1.40	1.00	1.46	1.00
Ave buy trade volume	22	15	22	15	23	15	21	15
Ave sell trade volume	337	25	22	15	366	43	768	68

Appendix Table A4

B. Weighted average spreads by credit rating and trading activity deciles: roundtrips starting from small buy trades

The table reports the median weighted average spread for dealer roundtrips starting from a buy trade of less than 100 bonds. Variable definitions are provided in Appendix Table A1. Reported spreads in this table exclude roundtrips with dealer buy transactions of less than 100 bonds. Spreads are reported for deciles of roundtrips based on trading activity variables in the 30 day period preceding the dealer buy transaction, where decile 1 is the lowest trading activity level group.

			All roundtrips (n=18,804,643)	A and above (n=9,966,888)	BBB (n=4,627,186)	BB (n=1,671,372)	B and lower (n=2,539,197)
	Trading activity decile:	High and low of trading activity variable within decile	Median spread	Median spread	Median spread	Median spread	Median spread
All			0.550	0.500	0.738	0.750	0.638
Trade count							
	1	0 - 16	0.750	0.800	0.750	0.656	0.500
	2	17 - 38	0.747	0.750	0.750	0.750	0.550
	3	39 - 69	0.660	0.600	0.721	0.790	0.625
	4	70 - 114	0.582	0.500	0.699	0.870	0.680
	5	115 - 179	0.500	0.394	0.660	0.899	0.700
	6-10	180 - 16,772	0.500	0.399	0.750	0.750	0.700
Days traded							
	1	0 - 6	0.750	0.762	0.743	0.633	0.500
	2	7 - 11	0.741	0.750	0.750	0.750	0.500
	3	12 - 15	0.667	0.625	0.727	0.776	0.550
	4	16 - 18	0.589	0.500	0.700	0.830	0.655
	5	19 - 19	0.500	0.400	0.715	0.800	0.700
	6-10	20 - 22	0.500	0.399	0.744	0.750	0.750

Appendix Table A4

C. Weighted average spreads by roundtrip type and counterparty type: roundtrips starting from small buy trades

	All roundtrips	Paired roundtrips			Other same day roundtrips				Roundtrips completed in more than one day			
		all	sell to customer	sell to dealer	all	sell to customer	sell to customer & dealer	sell to dealer	all	sell to customer	sell to customer & dealer	sell to dealer
All dealer buys	0.55 (18,804,643)	0.43 (9,670,872)	0.75 46.1%	0.20 53.9%	0.76 (2,185,768)	1.29 48.9%	0.86 6.2%	0.55 44.9%	0.88 (6,948,003)	1.04 43.8%	0.88 23.6%	0.77 32.7%
Buy from customer	0.75 (6,891,472)	0.50 (3,033,213)	1.70 46.1%	0.40 53.9%	1.00 (1,104,886)	2.00 48.9%	1.25 6.2%	0.50 44.9%	1.18 (2,753,373)	1.50 43.8%	1.27 23.6%	0.88 32.7%
Buy from dealer	0.50 (11,913,171)	0.40 (6,637,659)	0.64 46.1%	0.05 53.9%	0.67 (1,080,882)	0.80 48.9%	0.56 6.2%	0.60 44.9%	0.70 (4,194,630)	0.62 43.8%	0.56 23.6%	0.75 32.7%

Appendix Table A5

Explanation of variables included in regression switching models.

Variables are as defined in Appendix Table A1. The table provides the expected sign of the relationship to (1) the probability of completing a roundtrip the same day, and (2) roundtrip spreads.

		(1)	(2)	
Explanatory variables for:		Probability of completing roundtrip same day	Weighted ave roundtrips spreads	Explanation
Trading activity deciles (trade count or number of days traded)		+	+	proxy for search costs and/or expected time in inventory absent search
Bond specific characteristics				
	Rating	+	+	higher risk bonds (higher on rating scale) have higher inventory risk
	Time to maturity	+	+	longer maturity bonds have higher interest rate and therefore inventory risk
	144A bond	+	+	fewer buyers are available
	Bond age	+	+	bonds trade less often as they age
	Issue size	+/-	-	larger issues are traded more and by more dealers and customers
	# dealers trading the bond	+/-	-	search costs are lower when more dealers trade the bond
	Listed stock	-		dealer's ability to hedge bond in inventory
	Traded CDS	-		dealer's ability to hedge bond in inventory
Dealer characteristics				
	Top ten dealer (volume, trade count)	+	+/-	ability to manage/fund inventory; ability to locate counterparty;
	Dealer centrality	+	+/-	versus - core dealers are compensated for execution speed
	Dealer takes inventory (bond specific)	-		# days this dealer took this bond into inventory / # days dealer traded this bond
	Dealer traded in prior year (bond specific)	-		# days this dealer trades this bond in the prior year (-6 to -371)
Counterparty and trade characteristics				
	Buy volume >= 5000 or 1000-5000 vs. 500-1000	+	+	more inventory risk, longer to unwind
	# small (<100 bonds) sell trades/# sells		+	larger spreads on smaller trades
	Dealer buys from customer (vs. from dealer)	+/-	+/-	rents not shared in interdealer market; or, dealer assumes inventory risk
	Dealer sells to customer (vs. to dealer)		+	value of locating a customer
Market conditions			(included)	
	Quarter fixed effects			
	Crisis period indicator	+		
	Post crisis indicator	+		

Appendix Table A6

Endogenous Switching Regression Model: Comparisons for Decile 1

The table reports the results of the two stage endogenous switching regression model. Panel A provides comparison of decile 1 to all other deciles (2-10, the omitted group). Panel B omits deciles 6-10 and provides comparison of decile 1 to deciles 2-5 (the omitted group). Coefficients for additional included variables from the base specification (Tables 8 and 9) are not reported for brevity. *, **, and *** denote statistical significance at the 0.10, 0.05, and 0.01 level, respectively. Second stage regressions include quarter fixed effects. Robust standard errors clustered by issuer are shown in parentheses.

	Stage 1: Selection Equation (Probit)				Stage 2: Spread Model							
	<u>A & above</u>	<u>BBB</u>	<u>BB</u>	<u>B & below</u>	<u>A & above</u>		<u>BBB</u>		<u>BB</u>		<u>B & below</u>	
					same day	> 1 day	same day	> 1 day	same day	> 1 day	same day	> 1 day
<i>A. Sample includes all trade count deciles</i>												
Decile 1 (trade count)	0.1679*** (0.020)	0.1704*** (0.016)	0.2319*** (0.019)	0.1582*** (0.016)	0.0856*** (0.008)	0.1193*** (0.019)	0.0673*** (0.007)	0.0874*** (0.018)	0.0477** (0.024)	0.1371*** (0.028)	0.0653*** (0.006)	0.0119 (0.029)
Inverse Mills ratio					0.4920*** (0.017)	0.0443 (0.035)	0.3440*** (0.050)	-0.0734 (0.052)	0.139 (0.172)	-0.1020** (0.052)	0.0896*** (0.013)	-0.1100* (0.058)
Observed minus predicted from > 1 day					-0.092		-0.182		-0.035		-0.089	
Observed minus predicted from same day						-0.033		0.041		-0.038		0.067
Observations	1,374,972	955,421	511,009	1,258,738		1,374,972		955,421		511,009		1,258,738
<i>B. Sample omits trade count deciles 6-10</i>												
Decile 1 (trade count)	0.2460*** (0.015)	0.2522*** (0.016)	0.3203*** (0.018)	0.2771*** (0.017)	0.1131*** (0.008)	0.1608*** (0.021)	0.1036*** (0.008)	0.1201*** (0.025)	0.0787*** (0.015)	0.1579*** (0.038)	0.0607*** (0.006)	0.0399 (0.031)
Inverse Mills ratio					0.5030*** (0.012)	0.0372* (0.022)	0.4410*** (0.033)	-0.0766* (0.046)	0.2980*** (0.095)	-0.1070* (0.066)	0.0847*** (0.009)	-0.2010*** (0.066)
Observed minus predicted from > 1 day						-0.133		-0.247		-0.066		-0.097
Observed minus predicted from same day					-0.009		0.087		0.009		0.099	
Observations	2,915,822	1,813,435	809,892	2,040,304		2,915,822		1,813,435		809,892		2,040,304

Appendix Table A7

Endogenous Switching Regression Model: Impact of Dissemination Under TRACE

The sample includes all observations of dealer roundtrips from September 2003 to September 2004 (the pre-dissemination period) and February 2006 to February 2007 ("post" period). The period between is omitted due to the phase in of dissemination, under which trade information for less actively traded bonds is subject to delays (see timeline in Goldstein and Hotchkiss, 2009). Results are reported for bonds rated B and below. Additional control variables, not reported, are as included in Tables 7 and 8. Disseminated refers to the non-144A bonds which become disseminated in the post period.

	Stage 1: Selection Equation (Probit)	Stage 2: Spread Model	
		same day	> 1 day
Observations	553,826	553,826	
Decile 1	-0.0217 (0.036)	0.0486*** (0.017)	0.1087 (0.134)
Decile 2	-0.1823*** (0.031)	-0.0560*** (0.012)	-0.1302 (0.110)
Decile 3	-0.1696*** (0.029)	-0.0595*** (0.010)	-0.1201 (0.105)
Decile 4	-0.1204*** (0.025)	-0.0481*** (0.010)	-0.1071 (0.099)
Decile 5	-0.1073*** (0.023)	-0.0298*** (0.008)	-0.1639** (0.083)
Disseminated	-0.0466* (0.027)	0.0168 (0.014)	0.1949 (0.177)
Post	0.0891*** (0.028)	-0.0841*** (0.013)	0.3358 (0.213)
Disseminated*post	0.0208 (0.034)	-0.0346** (0.015)	-0.1480 (0.186)
Disseminated * post * decile 1	-0.1039*** (0.028)	-0.1403*** (0.013)	-0.5247*** (0.120)
Inverse Mills ratio		0.0712*** (0.009)	0.0116 (0.088)
Observed minus predicted from > 1 day		-0.242	
Observed minus predicted from same day			0.157

Table 1
Bond characteristics and trading activity

The table reports summary bond characteristics and monthly bond trading activity. The sample consists of 55,988 non-convertible bonds of 5,796 issuers with trades reported on TRACE between the July 2002 and March 2011 which are matched to Mergent and for which at least one dealer roundtrip is identified. Variables are defined in Appendix Table A1. Trading activity is averaged across months for each bond and excludes trades within 60 days of issuance and within 60 days of maturity or redemption.

	<u>Mean</u>	<u>Median</u>
<i>A. Bond and issuer characteristics</i>		
Issue size (\$000)	220,651	50,000
Time to maturity (years)	5.34	2.60
Age (years)	4.42	3.34
Rule 144A (1/0)	11.2%	0.0%
Rating	3.8	3.0
Stock listing: NYSE/AMEX/Nasdaq	70.3%	
OTC or other exchange	8.5%	
No traded stock	21.2%	
CDS traded	30.0%	
<i>B. Monthly bond trading activity</i>		
Monthly trade count	20.0	3.1
Monthly count of trades ≥ 100 bonds	6.3	0.6
Monthly volume (\$ in thousands)	14,315	717
Monthly number of non-zero trading days	3.4	1.3
Percent of months traded	49.3%	47.8%
<i>C. Trading activity and concentration of dealers</i>		
Number of dealers trading the bond	55.2	30.0
Average percentage interdealer trades	28.0%	27.7%
% of trade count by top ten dealers by trade count	41.1%	43.1%
% of volume by top ten dealers by volume	38.2%	34.0%
% of trade count by top 1% dealers by trade count	67.5%	70.7%
% of volume by top 1% dealers by volume	59.1%	61.9%

Table 2

Measures of bond trading activity by deciles and by credit rating groups

Table reports monthly trading activity by decile of the trading activity variable and by rating group, where decile 1 is the lowest trading activity group. Trading activity variables are defined in Appendix Table A1. For each bond, a monthly average of the trading variable is calculated excluding trades within 60 days of issuance and within 60 days of redemption or maturity. Mean and median of the monthly averages are reported.

Trading activity variable:	Decile	All ratings 55,988 bonds		A and above 31,748 bonds		BBB 10,023 bonds		BB 3,592 bonds		B and lower 10,625 bonds	
		Mean	Median	Mean	Median	Mean	Median	Mean	Median	Mean	Median
Trade count	all	20.0	3.1	17.5	2.2	29.3	5.9	28.4	10.7	16.0	3.2
	1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
	2	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
	3	0.8	0.8	0.8	0.8	0.9	0.9	0.8	0.8	0.8	0.8
	4	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
	5	2.5	2.4	2.5	2.4	2.5	2.4	2.5	2.5	2.5	2.5
	6-10	39.0	12.3	39.7	10.5	44.3	13.3	38.5	18.1	30.8	13.0
Days traded	all	3.4	1.3	2.9	0.9	4.6	2.5	5.4	4.0	3.1	1.2
	1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
	2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
	3	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
	4	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
	5	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	6-10	6.3	4.5	6.2	4.0	6.6	5.0	7.3	6.2	5.9	4.5

Table 3
Summary description of dealer roundtrips

The table provides description of 8,576,027 dealer roundtrips calculated for the dataset of 55,988 bonds, where the initial dealer buy is for at least 100 bonds. Variables are as defined in Appendix Table A1. Paired roundtrips are defined as a pair of trades where a given dealer buys and sells the same quantity of the same bond on the same day. Other roundtrips match each dealer buy of a given bond with one or more sell trades by that dealer in the same bond up to the face amount of the initial buy trade, using sell trades that occur within 60 days. Roundtrips which include any agency trades are excluded.

	All roundtrips (n=8,576,027)		Paired roundtrips (n=3,528,925; 41.1%)		Other same day roundtrips (n=1,463,295; 17.1%)		Roundtrips completed in more than one day (n=3,583,807; 41.8%)	
	<u>Mean</u>	<u>Median</u>	<u>Mean</u>	<u>Median</u>	<u>Mean</u>	<u>Median</u>	<u>Mean</u>	<u>Median</u>
Weighted ave holding period (days)	10.6	1.0	1.0	1.0	1.0	1.0	24.1	21.0
Length of holding period (days)	12.8	1.0	1.0	1.0	1.0	1.0	29.3	28.0
Time to last sell (seconds)	1,021,322	6,540	1,515	99	5,593	2,380	2,440,241	2,340,822
# sells in the roundtrip	2.13	1.00	1.00	1.00	2.08	1.00	3.27	2.00
Ave buy trade volume (# bonds)	2,079	795	1,760	600	2,415	1,000	2,256	820
Ave sell trade volume (# bonds)	2,611	1,000	1,760	600	3,333	1,500	3,155	1,188

Table 4
Spreads on dealer roundtrips

The table reports the median weighted average spread for dealer roundtrips. Variable definitions are provided in Appendix Table A1. Reported spreads in this table exclude roundtrips with dealer buy transactions of less than 100 bonds. Spreads are reported for deciles of roundtrips based on trading activity variables in the 30 day period preceding the dealer buy transaction, where decile 1 is the lowest trading activity level group. a,b,c for deciles 1 through 5 denote the median is significantly different from that of deciles 6-10 at the 1 percent, 5 percent, or 10 percent level, respectively.

A. Weighted ave spreads for dealer roundtrips by credit rating and trading activity deciles

			<u>All roundtrips (n=8,576,027)</u>	<u>A and above (n=3,375,655)</u>	<u>BBB (n=2,070,371)</u>	<u>BB (n=898,550)</u>	<u>B and lower (n=2,231,451)</u>
	Trading activity decile:	High and low of trading activity variable within decile	Median spread	Median spread	Median spread	Median spread	Median spread
All			0.150	0.106	0.147	0.220	0.250
Trade count							
	1	0 - 16	0.179 ^a	0.100 ^a	0.147 ^a	0.250	0.250 ^a
	2	17 - 38	0.159 ^a	0.102 ^a	0.125 ^a	0.250 ^a	0.250 ^a
	3	39 - 69	0.133 ^a	0.100 ^a	0.110 ^a	0.188 ^a	0.250 ^a
	4	70 - 114	0.125 ^a	0.094 ^a	0.109 ^a	0.188 ^a	0.250 ^a
	5	115 - 179	0.125 ^a	0.087 ^a	0.123 ^a	0.188 ^a	0.250 ^a
	6-10	180 - 16,772	0.162	0.125	0.200	0.240	0.250
Days traded							
	1	0 - 6	0.213 ^a	0.125 ^a	0.181	0.250 ^a	0.250 ^a
	2	7 - 11	0.188 ^a	0.105 ^a	0.132 ^a	0.250	0.250 ^a
	3	12 - 15	0.151	0.100 ^a	0.125 ^a	0.200 ^a	0.250 ^a
	4	16 - 18	0.126 ^a	0.100 ^a	0.125 ^a	0.188 ^a	0.250 ^a
	5	19 - 19	0.125 ^a	0.100 ^a	0.136 ^a	0.188 ^a	0.220 ^a
	6-10	20 - 22	0.137	0.110	0.172	0.200	0.213

Table 4 - continued

B. Weighted ave spreads for dealer roundtrips by counterparty type

	All dealer buys		Buy from customer		Buy from dealer	
	median spread	(n) ; % of n	median spread	(n) ; % of n	median spread	(n) ; % of n
All roundtrips						
all	0.15	(8,576,027)	0.24	(3,971,759)	0.13	(4,604,268)
sell to customer	0.25	45.6%	0.25	54.4%	0.25	38.0%
sell to customer & dealer	0.31	10.9%	0.30	13.8%	0.33	8.4%
sell to dealer	0.07	43.5%	0.18	31.8%	0.06	53.6%
Paired roundtrips						
all	0.10	(3,528,925)	0.13	(1,359,073)	0.08	(2,169,852)
sell to customer	0.19	46.1%	0.13	55.6%	0.22	40.2%
sell to dealer	0.06	53.9%	0.13	44.4%	0.06	59.8%
Other same day roundtrips						
all	0.23	(1,463,295)	0.25	(595,141)	0.13	(868,154)
sell to customer	0.29	48.9%	0.25	61.4%	0.32	40.3%
sell to customer & dealer	0.26	6.2%	0.30	8.8%	0.23	4.4%
sell to dealer	0.08	44.9%	0.25	29.8%	0.06	55.3%
Roundtrips completed in more than one day						
all	0.29	(3,583,807)	0.25	(2,017,545)	0.35	(1,566,262)
sell to customer	0.30	43.8%	0.25	51.6%	0.41	33.7%
sell to customer & dealer	0.32	23.6%	0.30	24.5%	0.35	22.4%
sell to dealer	0.26	32.7%	0.21	23.9%	0.31	43.9%

Table 5
Time for completion of roundtrips

The table reports the total number of roundtrips by rating category and by trading activity deciles, and the percentage of roundtrips which are paired roundtrips, other same day roundtrips, and roundtrips completed in more than one day. All variables are as defined in Appendix Table A1. The table includes 8,576,027 roundtrips starting from a dealer buy of at least 100 bonds and excludes roundtrips that include any agency trades.

		Percent of total # roundtrips:			Weighted ave holding period (days) for roundtrips completed in more than one day
		Total #	Paired roundtrips	Other same day roundtrips	
All roundtrips		8,576,027	41.1%	17.1%	41.8%
Rating group					
A and above		3,375,655	41.1%	14.1%	44.7%
BBB		2,070,371	40.1%	16.6%	43.3%
BB		898,550	39.4%	18.7%	41.9%
B and below		2,231,451	42.9%	21.2%	35.9%
Trade count					
Deciles:	1	905,501	49.9%	12.2%	38.0%
	2	1,004,219	41.7%	13.9%	44.4%
	3	1,028,011	40.0%	15.2%	44.8%
	4	997,979	39.7%	16.2%	44.1%
	5	958,228	39.8%	16.8%	43.4%
	6-10	3,682,089	39.9%	19.9%	40.1%
Days traded					
Deciles:	1	820,287	52.9%	12.6%	34.5%
	2	931,650	43.4%	14.2%	42.4%
	3	1,027,692	40.9%	15.7%	43.5%
	4	1,221,131	40.4%	17.1%	42.5%
	5	892,266	41.5%	18.9%	39.6%
	6-10	3,683,001	38.2%	18.7%	43.1%

Table 6

Roundtrip completion for subsamples of bonds by rating and by dealer characteristics

The table reports the total number of roundtrips by rating category and by trading activity deciles, and the percentage of roundtrips which are paired roundtrips, other same day roundtrips, and roundtrips completed in more than one day. Panel A reports these statistics for subgroups of the highest and lowest rated bonds. Panel B reports roundtrip completions based on whether the dealer is a *top ten: volume* dealer and on whether more than 50% of the dealer's trades on TRACE are for less than 100 bonds (*small trade dealer*). All variables are as defined in Appendix Table A1. The table includes all roundtrips starting from a dealer buy of at least 100 bonds and excludes roundtrips that include any agency trades.

A. Rating category subsamples

		A rated and above					B rated and below				
		Total	Percent of total				Total	Percent of total			
		# roundtrips	Paired roundtrips	Other same day roundtrips	Roundtrips completed in more than one day	Weighted ave holding period for roundtrips completed in more than one day (median days)	# roundtrips	Paired roundtrips	Other same day roundtrips	Roundtrips completed in more than one day	Weighted ave holding period for roundtrips completed in more than one day (median days)
Trade count											
Deciles:	1	362,366	48.7%	9.0%	42.3%	14.0	223,906	53.1%	17.3%	29.7%	24.9
	2	327,320	42.0%	10.4%	47.6%	16.0	293,565	43.4%	18.2%	38.4%	24.0
	3	325,033	40.8%	11.4%	47.8%	18.2	319,477	41.2%	19.4%	39.4%	23.2
	4	344,220	39.8%	12.6%	47.5%	18.2	306,629	41.3%	20.3%	38.4%	22.9
	5	376,917	39.6%	13.4%	46.9%	22.0	271,969	41.4%	21.5%	37.1%	22.0
	6-10	1,639,799	40.0%	17.0%	43.0%	21.9	815,905	41.6%	24.4%	34.0%	20.0
Days traded											
Deciles:	1	327,817	51.1%	9.3%	39.6%	12.0	212,763	56.9%	17.9%	25.2%	22.5
	2	296,399	43.2%	10.4%	46.4%	15.0	289,422	45.4%	18.7%	35.9%	23.0
	3	292,868	41.5%	11.4%	47.0%	17.0	353,184	42.1%	19.7%	38.2%	23.0
	4	380,457	41.6%	13.2%	45.2%	20.2	407,356	40.9%	21.2%	37.9%	22.5
	5	366,015	43.1%	15.9%	41.0%	22.0	227,208	40.8%	23.1%	36.1%	22.0
	6-10	1,712,099	38.3%	16.0%	45.7%	22.0	741,518	39.9%	23.4%	36.7%	21.0

Table 6 - continued

B. Top ten (volume) dealers versus others

		Top ten dealer: volume					Other dealers				
		Total	Percent of total			Weighted ave holding period for roundtrips completed in more than one day (median days)	Total	Percent of total			Weighted ave holding period for roundtrips completed in more than one day (median days)
		# roundtrips	Paired roundtrips	Other same day roundtrips	Roundtrips completed in more than one day		# roundtrips	Paired roundtrips	Other same day roundtrips	Roundtrips completed in more than one day	
Trade count deciles:	1	542,802	37.8%	12.6%	49.6%	21.3	362,699	68.0%	11.4%	20.6%	8.0
	2	655,357	31.7%	14.6%	53.8%	22.7	348,862	60.5%	12.8%	26.7%	11.0
	3	681,395	30.9%	15.9%	53.2%	23.0	346,616	57.8%	13.8%	28.3%	13.6
	4	657,734	31.2%	16.8%	52.0%	23.0	340,245	56.2%	15.0%	28.8%	15.0
	5	617,223	31.5%	17.3%	51.2%	23.1	341,005	54.8%	15.8%	29.4%	16.7
	6-10	2,164,047	32.9%	20.1%	46.9%	22.1	1,518,042	49.9%	19.6%	30.5%	17.9
Days traded deciles:	1	459,373	41.3%	13.0%	45.6%	20.2	360,914	67.6%	12.2%	20.3%	7.0
	2	593,473	33.1%	14.7%	52.2%	22.0	338,177	61.5%	13.4%	25.1%	10.0
	3	681,637	31.9%	16.2%	51.9%	22.8	346,055	58.6%	14.5%	26.9%	13.0
	4	806,910	32.1%	17.5%	50.3%	23.0	414,221	56.4%	16.2%	27.4%	15.0
	5	552,126	33.1%	19.2%	47.7%	23.9	340,140	55.2%	18.3%	26.5%	18.0
	6-10	2,225,039	31.0%	18.9%	50.1%	22.9	1,457,962	49.2%	18.4%	32.4%	18.0

Table 7
First Stage Probit Model

The table provides probit models for the selection of roundtrips completed within the same day versus longer roundtrips where bonds are taken into overnight dealer inventory. The dependent variable equals one for paired roundtrips and other same day roundtrips, and zero otherwise. Other variables are as defined in Appendix Table A1. *, **, and *** denote statistical significance at the 0.10, 0.05, and 0.01 level, respectively. Standard errors clustered by issuer are shown in parentheses.

VARIABLES	Trading activity variable:				Days traded			
	A & above	BBB	BB	B & below	A & above	BBB	BB	B & below
Decile 1	0.1158*** (0.030)	0.0430 (0.030)	0.1696*** (0.038)	0.0067 (0.031)	0.3925*** (0.021)	0.3394*** (0.029)	0.3981*** (0.028)	0.3186*** (0.027)
Decile 2	-0.0243 (0.019)	-0.1120*** (0.024)	-0.0223 (0.029)	-0.1486*** (0.025)	0.1826*** (0.015)	0.1291*** (0.023)	0.1428*** (0.021)	0.0768*** (0.019)
Decile 3	-0.0512*** (0.015)	-0.1389*** (0.020)	-0.0884*** (0.025)	-0.1639*** (0.022)	0.1183*** (0.012)	0.0640*** (0.020)	0.0481*** (0.018)	0.0199 (0.016)
Decile 4	-0.0637*** (0.012)	-0.1258*** (0.017)	-0.0779*** (0.021)	-0.1378*** (0.018)	0.1033*** (0.009)	0.0563*** (0.013)	0.0324** (0.014)	0.0113 (0.013)
Decile 5	-0.0620*** (0.010)	-0.1027*** (0.013)	-0.0608*** (0.018)	-0.1049*** (0.015)	0.1464*** (0.008)	0.1092*** (0.012)	0.0652*** (0.013)	0.0387*** (0.010)
Time to maturity	-0.0034*** (0.001)	-0.0011 (0.001)	0.0007 (0.001)	-0.0008 (0.001)	-0.0039*** (0.001)	-0.0009 (0.001)	0.0006 (0.001)	-0.0011 (0.001)
Rule 144A bond	0.1616*** (0.037)	0.0860*** (0.030)	0.1021*** (0.030)	0.0453** (0.023)	0.1609*** (0.037)	0.1209*** (0.032)	0.1293*** (0.032)	0.0797*** (0.024)
Bond age	0.0202*** (0.002)	0.0136*** (0.003)	0.0130*** (0.003)	0.0167*** (0.003)	0.0154*** (0.002)	0.0088*** (0.003)	0.0102*** (0.002)	0.0125*** (0.003)
Issue size	0.0135 (0.009)	0.0317*** (0.009)	0.1114*** (0.017)	0.0206 (0.013)	0.0256*** (0.008)	0.0400*** (0.009)	0.1207*** (0.020)	0.0364*** (0.013)
Ln(# dealers)	0.0633** (0.025)	0.0356** (0.018)	0.0336** (0.016)	0.0195 (0.018)	0.1359*** (0.021)	0.1215*** (0.019)	0.0827*** (0.015)	0.0871*** (0.019)
Listed stock	0.0038 (0.032)	-0.0318 (0.032)	-0.0678** (0.029)	-0.0537*** (0.018)	-0.0042 (0.029)	-0.0260 (0.033)	-0.0634** (0.027)	-0.0557*** (0.020)
CDS traded	0.0225 (0.020)	-0.0360 (0.027)	-0.0996*** (0.023)	-0.1019*** (0.019)	0.0274 (0.020)	-0.0294 (0.029)	-0.0954*** (0.023)	-0.0877*** (0.020)
Top ten dealer: volume	-0.2912*** (0.017)	-0.3816*** (0.018)	-0.5062*** (0.018)	-0.4787*** (0.011)	-0.2924*** (0.017)	-0.3836*** (0.018)	-0.5085*** (0.018)	-0.4821*** (0.011)
Top ten dealer: trade count	0.1742*** (0.019)	0.1136*** (0.013)	-0.0374 (0.026)	-0.0177 (0.019)	0.1715*** (0.019)	0.1104*** (0.013)	-0.0385 (0.026)	-0.0187 (0.019)
Eigenvector centrality	-18.6567*** (0.319)	-15.9960*** (0.281)	-8.6268*** (0.403)	-7.6905*** (0.282)	-18.7746*** (0.319)	-16.1347*** (0.287)	-8.6898*** (0.407)	-7.8074*** (0.289)
Dealer takes inventory (this bond)	-1.5285*** (0.034)	-1.5388*** (0.064)	-1.5719*** (0.049)	-1.5988*** (0.041)	-1.5315*** (0.033)	-1.5509*** (0.065)	-1.5785*** (0.049)	-1.6063*** (0.041)
Dealer traded in prior year (this bond)	-0.0010*** (0.000)	-0.0014*** (0.000)	-0.0023*** (0.000)	-0.0014*** (0.000)	-0.0008*** (0.000)	-0.0012*** (0.000)	-0.0021*** (0.000)	-0.0012*** (0.000)
Buy volume >= 5000	-0.0764*** (0.014)	0.1230*** (0.014)	0.3073*** (0.027)	0.3886*** (0.023)	-0.0699*** (0.014)	0.1177*** (0.014)	0.3068*** (0.025)	0.3872*** (0.022)
1000 <= buy volume < 5000	-0.1353*** (0.014)	0.0213** (0.010)	0.1368*** (0.021)	0.2366*** (0.016)	-0.1315*** (0.013)	0.0193* (0.010)	0.1396*** (0.019)	0.2427*** (0.015)
500 <= buy volume < 1000	-0.1353*** (0.009)	-0.0974*** (0.010)	-0.0431*** (0.012)	0.0427*** (0.014)	-0.1321*** (0.009)	-0.0975*** (0.010)	-0.0406*** (0.012)	0.0439*** (0.013)
Buy from customer	-0.2638*** (0.009)	-0.1397*** (0.016)	-0.0551** (0.025)	-0.0032 (0.011)	-0.2686*** (0.010)	-0.1477*** (0.017)	-0.0612** (0.026)	-0.0164 (0.012)
Crisis	0.2279*** (0.012)	0.2488*** (0.018)	0.1180*** (0.017)	0.1019*** (0.013)	0.2347*** (0.011)	0.2494*** (0.018)	0.1123*** (0.018)	0.0815*** (0.014)
Post-crisis	-0.0585*** (0.016)	-0.0515*** (0.017)	-0.0909*** (0.018)	-0.0695*** (0.025)	-0.0327** (0.015)	-0.0151 (0.018)	-0.0694*** (0.018)	-0.0728*** (0.026)
Constant	2.1069*** (0.147)	2.1246*** (0.092)	1.6047*** (0.096)	1.7084*** (0.090)	1.6234*** (0.123)	1.5456*** (0.093)	1.2601*** (0.090)	1.2608*** (0.094)
Observations	2,915,822	1,813,435	809,892	2,040,304	2,915,822	1,813,435	809,892	2,040,304
Pseudo. R-squared	0.21	0.19	0.17	0.15	0.21	0.19	0.17	0.15

Table 8
Second Stage Endogenous Switching Regression Models

The dependent variable is the weighted average spread for dealer roundtrips completed the same day or completed over longer holding periods. Regressions include quarter fixed effects. Robust standard errors (in parentheses) cluster by issuer. Inverse Mills ratio is calculated from Stage 1 models reported in Table 8. Variables are as defined in Appendix Table A1.

A. Using indicators for deciles by trade count

Roundtrip is:	A & above		BBB		BB		B & below	
	same day	> 1 day	same day	> 1 day	same day	> 1 day	same day	> 1 day
Decile 1	0.2408*** (0.032)	0.1000*** (0.015)	-0.0026 (0.056)	0.0444** (0.020)	0.0020 (0.077)	0.0365* (0.022)	-0.0791 (0.073)	0.0330*** (0.011)
Decile 2	0.1365*** (0.025)	0.0145 (0.013)	-0.0904* (0.053)	-0.0402** (0.018)	-0.1257* (0.066)	-0.0290* (0.016)	-0.1374** (0.063)	-0.0215** (0.009)
Decile 3	0.0410* (0.022)	-0.0161 (0.012)	-0.1305*** (0.049)	-0.0656*** (0.016)	-0.1500** (0.063)	-0.0492*** (0.014)	-0.1145** (0.057)	-0.0354*** (0.008)
Decile 4	0.0092 (0.018)	-0.0322*** (0.009)	-0.1187*** (0.046)	-0.0694*** (0.013)	-0.1393** (0.058)	-0.0462*** (0.013)	-0.0812 (0.053)	-0.0279*** (0.007)
Decile 5	-0.0018 (0.014)	-0.0353*** (0.007)	-0.1222*** (0.043)	-0.0635*** (0.011)	-0.1661*** (0.056)	-0.0447*** (0.012)	-0.0759* (0.043)	-0.0203*** (0.006)
Time to maturity	0.0175*** (0.001)	0.0090*** (0.001)	0.0159*** (0.001)	0.0069*** (0.001)	0.0151*** (0.003)	0.0057*** (0.001)	0.0047 (0.004)	0.0014* (0.001)
Rule 144A bond	0.0736** (0.032)	0.0341** (0.015)	0.1076** (0.047)	0.1325*** (0.017)	0.0424 (0.067)	0.0735*** (0.015)	-0.0019 (0.050)	0.0624*** (0.009)
Bond age	0.0020 (0.003)	0.0045*** (0.001)	0.0056 (0.004)	0.0055*** (0.002)	0.0150* (0.008)	0.0041* (0.002)	0.0454*** (0.007)	0.0105*** (0.001)
Issue size	-0.0651*** (0.013)	-0.0258*** (0.004)	-0.1323*** (0.022)	-0.0697*** (0.011)	-0.0242 (0.048)	-0.0658*** (0.009)	-0.0514* (0.027)	-0.0433*** (0.006)
Ln(# dealers)	0.0818*** (0.018)	0.0445*** (0.008)	0.0345 (0.031)	0.0806*** (0.008)	-0.0107 (0.053)	0.0525*** (0.012)	-0.0369 (0.035)	0.0124* (0.007)
Top ten dealer: volume	-0.0587*** (0.011)	-0.1513*** (0.008)	-0.0953*** (0.014)	-0.2094*** (0.014)	-0.1604*** (0.035)	-0.1532*** (0.038)	-0.1478*** (0.028)	-0.0640*** (0.008)
Top ten dealer: trade count	0.1794*** (0.011)	0.1525*** (0.008)	0.1727*** (0.015)	0.0938*** (0.008)	0.0787*** (0.021)	0.0350*** (0.013)	0.1207*** (0.021)	0.0759*** (0.012)
Eigenvector centrality	-1.4480*** (0.389)	-5.7738*** (0.201)	-3.0683*** (0.668)	-4.3888*** (0.333)	-0.0212 (0.599)	-1.5445*** (0.475)	-1.0959** (0.541)	-0.5395*** (0.194)
Buy volume >= 5000	-0.2478*** (0.013)	-0.1949*** (0.009)	-0.3631*** (0.017)	-0.2187*** (0.014)	-0.3090*** (0.045)	-0.2039*** (0.022)	-0.2509*** (0.030)	-0.2174*** (0.011)
1000 <= buy volume < 5000	-0.2074*** (0.008)	-0.1781*** (0.008)	-0.2584*** (0.017)	-0.2187*** (0.010)	-0.3103*** (0.039)	-0.2159*** (0.020)	-0.3260*** (0.024)	-0.2200*** (0.009)
500 <= buy volume < 1000	-0.1253*** (0.008)	-0.1271*** (0.006)	-0.1665*** (0.012)	-0.1677*** (0.007)	-0.2078*** (0.033)	-0.1824*** (0.015)	-0.2318*** (0.018)	-0.1726*** (0.007)
Ratio small/total # sells	0.3386*** (0.013)	0.3859*** (0.019)	0.3837*** (0.035)	0.6929*** (0.047)	0.5562*** (0.038)	0.8039*** (0.058)	0.5301*** (0.042)	0.9273*** (0.029)
Buy from customer	0.0519*** (0.009)	-0.0525*** (0.006)	0.0130 (0.017)	-0.0002 (0.007)	0.0259 (0.019)	0.0466*** (0.010)	-0.0541*** (0.016)	0.0712*** (0.006)
% sale volume to customers	0.1887*** (0.010)	0.1130*** (0.008)	0.2218*** (0.022)	0.1733*** (0.016)	0.2381*** (0.021)	0.1395*** (0.011)	0.2342*** (0.017)	0.1456*** (0.007)
Constant	0.9046*** (0.146)	0.1596*** (0.055)	1.0936*** (0.421)	0.1562* (0.088)	0.6689 (0.491)	0.3243*** (0.080)	0.5577** (0.276)	0.4947*** (0.061)
Inverse Mills ratio	0.0368* (0.022)	0.5040*** (0.012)	-0.0783* (0.044)	0.4460*** (0.031)	-0.1110* (0.065)	0.3090*** (0.077)	-0.1920*** (0.062)	0.0884*** (0.010)
Observations	2,915,822		1,813,435		809,892		2,040,304	
p_chi ²	0.098	0.000	0.071	0.000	0.086	0.000	0.002	0.000
Observed minus predicted from > 1 day	-0.132		-0.250		-0.067		-0.099	
Observed minus predicted from same day		-0.009		0.088		0.010		0.100

Table 8 - continued

B. Using indicators for deciles by days traded

Roundtrip is:	A & above		BBB		BB		B & below	
	same day	> 1 day	same day	> 1 day	same day	> 1 day	same day	> 1 day
Decile 1	0.2313*** (0.012)	0.3063*** (0.032)	0.1683*** (0.017)	0.1840*** (0.047)	0.1189*** (0.028)	0.2221*** (0.071)	0.1023*** (0.010)	0.1699*** (0.056)
Decile 2	0.1160*** (0.010)	0.2011*** (0.023)	0.0578*** (0.015)	0.0469 (0.043)	0.0416*** (0.014)	0.1005** (0.049)	0.0269*** (0.008)	0.0864* (0.048)
Decile 3	0.0643*** (0.009)	0.0979*** (0.019)	0.0170 (0.013)	-0.0248 (0.039)	0.0021 (0.011)	0.0015 (0.047)	-0.0015 (0.007)	0.0855* (0.044)
Decile 4	0.0393*** (0.006)	0.0393*** (0.013)	0.0032 (0.010)	-0.0192 (0.037)	-0.0013 (0.009)	-0.0395 (0.041)	-0.0073 (0.005)	0.0583 (0.038)
Decile 5	0.0407*** (0.003)	0.0153* (0.008)	0.0250*** (0.006)	0.0211 (0.031)	0.0105 (0.008)	-0.0404 (0.029)	-0.0005 (0.003)	0.0834*** (0.031)
Time to maturity	0.0088*** (0.001)	0.0173*** (0.001)	0.0070*** (0.001)	0.0160*** (0.001)	0.0057*** (0.001)	0.0150*** (0.003)	0.0013* (0.001)	0.0044 (0.004)
Rule 144A bond	0.0357** (0.015)	0.0727** (0.032)	0.1466*** (0.018)	0.1300*** (0.046)	0.0842*** (0.016)	0.0742 (0.068)	0.0721*** (0.009)	0.0293 (0.047)
Bond age	0.0020 (0.001)	0.0008 (0.003)	0.0033** (0.002)	0.0028 (0.004)	0.0029 (0.002)	0.0119 (0.008)	0.0092*** (0.001)	0.0423*** (0.007)
Issue size	-0.0198*** (0.004)	-0.0623*** (0.012)	-0.0656*** (0.010)	-0.1244*** (0.020)	-0.0608*** (0.009)	-0.0048 (0.051)	-0.0404*** (0.006)	-0.0307 (0.027)
Ln(# dealers)	0.0823*** (0.007)	0.0977*** (0.018)	0.1193*** (0.008)	0.0864*** (0.031)	0.0750*** (0.012)	0.0529 (0.056)	0.0303*** (0.006)	0.0277 (0.029)
Top ten dealer: volume	-0.1510*** (0.008)	-0.0586*** (0.011)	-0.2073*** (0.015)	-0.0961*** (0.015)	-0.1476*** (0.045)	-0.1653*** (0.035)	-0.0609*** (0.007)	-0.1554*** (0.029)
Top ten dealer: trade count	0.1526*** (0.008)	0.1795*** (0.011)	0.0944*** (0.009)	0.1732*** (0.015)	0.0372** (0.015)	0.0803*** (0.021)	0.0767*** (0.012)	0.1226*** (0.021)
Eigenvector centrality	-5.7577*** (0.200)	-1.4476*** (0.394)	-4.3620*** (0.346)	-3.1336*** (0.706)	-1.4895*** (0.561)	-0.0474 (0.617)	-0.5337*** (0.195)	-1.2910** (0.553)
Buy volume >= 5000	-0.1925*** (0.009)	-0.2469*** (0.013)	-0.2226*** (0.013)	-0.3616*** (0.017)	-0.2073*** (0.021)	-0.3038*** (0.044)	-0.2184*** (0.011)	-0.2377*** (0.030)
1000 <= buy volume < 5000	-0.1766*** (0.007)	-0.2060*** (0.008)	-0.2206*** (0.010)	-0.2565*** (0.017)	-0.2166*** (0.019)	-0.3054*** (0.038)	-0.2183*** (0.009)	-0.3142*** (0.024)
500 <= buy volume < 1000	-0.1257*** (0.006)	-0.1240*** (0.008)	-0.1682*** (0.006)	-0.1653*** (0.012)	-0.1820*** (0.014)	-0.2052*** (0.031)	-0.1724*** (0.007)	-0.2281*** (0.018)
Ratio small/total # sells	0.3886*** (0.019)	0.3380*** (0.012)	0.6988*** (0.048)	0.3954*** (0.034)	0.8125*** (0.063)	0.5715*** (0.037)	0.9303*** (0.029)	0.5476*** (0.042)
Buy from customer	-0.0541*** (0.006)	0.0536*** (0.009)	-0.0022 (0.008)	0.0087 (0.018)	0.0449*** (0.010)	0.0185 (0.020)	0.0681*** (0.007)	-0.0662*** (0.017)
% sale volume to customers	0.1139*** (0.008)	0.1908*** (0.010)	0.1745*** (0.016)	0.2194*** (0.022)	0.1400*** (0.011)	0.2338*** (0.021)	0.1432*** (0.007)	0.2250*** (0.017)
Constant	-0.1279*** (0.046)	0.7782*** (0.143)	-0.1168 (0.085)	0.7552* (0.412)	0.1674* (0.088)	0.2597 (0.483)	0.3704*** (0.062)	0.1587 (0.245)
Inverse Mills ratio	0.5020*** (0.012)	0.0357* (0.022)	0.4400*** (0.033)	-0.0770* (0.047)	0.2970*** (0.094)	-0.1110* (0.068)	0.0872*** (0.009)	-0.2050*** (0.065)
Observations	2,915,822		1,813,435		809,892		2,040,304	
p_chi ²	0.000	0.109	0.000	0.096	0.003	0.102	0.000	0.002
Observed minus predicted from > 1 day	-0.135		-0.249		-0.068		-0.098	
Observed minus predicted from same day		-0.008		0.088		0.009		0.101

Table 9
Second Stage Endogenous Switching Regression Models: Effect of Market Stress

The dependent variable is the weighted average spread for dealer roundtrips completed the same day or completed over longer holding periods. Panel A includes interactions of trading activity deciles with indicator for the financial crisis period, defined as the second quarter of 2007 through the second quarter of 2009. Panel B includes interactions with an indicator for days of increases in the VIX in the 95th percentile of VIX changes over our sample period. Regressions include quarter fixed effects. Robust standard errors (in parentheses) cluster by issuer. Inverse Mills ratio is calculated from Stage 1 models not reported. Additional control variables not reported are as specified in Table 9. Variables are as defined in Appendix Table A1.

A. Interactions of crisis with deciles by trade count

Roundtrip is:	A & above		BBB		BB		B & below	
	same day	> 1 day	same day	> 1 day	same day	> 1 day	same day	> 1 day
Decile 1	0.1016*** (0.017)	0.1957*** (0.032)	0.0371* (0.022)	-0.0372 (0.060)	0.0480** (0.023)	-0.0993 (0.075)	0.0462*** (0.012)	-0.1809** (0.078)
Decile 2	0.0178 (0.015)	0.0936*** (0.027)	-0.0480** (0.020)	-0.1138** (0.057)	-0.0290 (0.018)	-0.1880*** (0.065)	-0.0135 (0.010)	-0.2105*** (0.069)
Decile 3	-0.0148 (0.013)	0.0142 (0.024)	-0.0731*** (0.018)	-0.1444*** (0.054)	-0.0496*** (0.015)	-0.1826*** (0.063)	-0.0317*** (0.009)	-0.1857*** (0.064)
Decile 4	-0.0305*** (0.010)	-0.0223 (0.020)	-0.0755*** (0.015)	-0.1239** (0.050)	-0.0453*** (0.015)	-0.1529** (0.061)	-0.0253*** (0.008)	-0.1104* (0.060)
Decile 5	-0.0354*** (0.008)	-0.0291* (0.016)	-0.0671*** (0.012)	-0.1234*** (0.045)	-0.0473*** (0.012)	-0.1628*** (0.057)	-0.0182*** (0.007)	-0.0992* (0.051)
Decile 1 * crisis	-0.0057 (0.015)	0.1854*** (0.044)	0.0317 (0.024)	0.1902*** (0.068)	-0.0429* (0.024)	0.4772* (0.249)	-0.0531*** (0.018)	0.5038*** (0.146)
Decile 2 * crisis	-0.0141 (0.012)	0.2125*** (0.044)	0.0384 (0.024)	0.1534** (0.071)	0.0006 (0.026)	0.3414 (0.252)	-0.0338** (0.016)	0.3866*** (0.146)
Decile 3 * crisis	-0.0051 (0.012)	0.1307*** (0.040)	0.0408* (0.025)	0.1046 (0.075)	0.0028 (0.022)	0.2102 (0.256)	-0.0172 (0.015)	0.3829** (0.150)
Decile 4 * crisis	-0.0077 (0.013)	0.1625*** (0.039)	0.0350 (0.024)	0.0503 (0.072)	-0.0045 (0.022)	0.1063 (0.245)	-0.0135 (0.015)	0.1817 (0.152)
Decile 5 * crisis	0.0004 (0.012)	0.1387*** (0.034)	0.0227 (0.022)	0.0239 (0.078)	0.0146 (0.018)	-0.0005 (0.251)	-0.0111 (0.013)	0.1507 (0.135)
Inverse Mills ratio	0.5040*** (0.012)	0.0421* (0.022)	0.4460*** (0.031)	-0.0757* (0.044)	0.3080*** (0.078)	-0.1070* (0.064)	0.0885*** (0.010)	-0.1910*** (0.062)
Observations	2,915,822		1,813,435		809,892		2,040,304	
p_chi ²	0.000	0.062	0.000	0.077	0.000	0.098	0.000	0.002
Observed minus predicted from > 1 day	-0.132		-0.249		-0.068		-0.098	
Observed minus predicted from same day		-0.009		0.088		0.010		0.100

Table 9 - continued

B. Interactions of high delta vix with deciles by trade count

Roundtrip is:	A & above		BBB		BB		B & below	
	same day	> 1 day	same day	> 1 day	same day	> 1 day	same day	> 1 day
Decile 1	0.1005*** (0.015)	0.2420*** (0.032)	0.0409** (0.021)	-0.0195 (0.055)	0.0380* (0.023)	-0.0077 (0.077)	0.0344*** (0.011)	-0.0878 (0.074)
Decile 2	0.0156 (0.013)	0.1325*** (0.026)	-0.0423** (0.019)	-0.1034** (0.052)	-0.0289* (0.017)	-0.1272* (0.066)	-0.0217** (0.009)	-0.1372** (0.063)
Decile 3	-0.0148 (0.012)	0.0371* (0.022)	-0.0672*** (0.016)	-0.1459*** (0.048)	-0.0494*** (0.015)	-0.1586** (0.063)	-0.0345*** (0.008)	-0.1148** (0.057)
Decile 4	-0.0313*** (0.009)	0.0048 (0.018)	-0.0703*** (0.014)	-0.1302*** (0.045)	-0.0463*** (0.014)	-0.1409** (0.058)	-0.0281*** (0.007)	-0.0793 (0.054)
Decile 5	-0.0348*** (0.007)	-0.0077 (0.015)	-0.0645*** (0.011)	-0.1350*** (0.042)	-0.0447*** (0.013)	-0.1593*** (0.056)	-0.0198*** (0.006)	-0.0767* (0.044)
Decile 1 * high delta VIX	-0.0020 (0.015)	-0.0444 (0.043)	0.0698*** (0.018)	0.3209*** (0.100)	-0.0217 (0.028)	0.2172* (0.112)	-0.0249 (0.019)	0.2029* (0.123)
Decile 2 * high delta VIX	-0.0164 (0.011)	0.0719* (0.039)	0.0390** (0.018)	0.2319** (0.100)	0.0016 (0.034)	0.0211 (0.089)	0.0036 (0.018)	-0.0045 (0.123)
Decile 3 * high delta VIX	-0.0211* (0.011)	0.0677 (0.042)	0.0289* (0.017)	0.2943*** (0.101)	0.0070 (0.026)	0.1963** (0.091)	-0.0170 (0.016)	0.0085 (0.121)
Decile 4 * high delta VIX	-0.0126 (0.009)	0.0770 (0.047)	0.0159 (0.016)	0.2136** (0.098)	0.0015 (0.023)	0.0357 (0.096)	0.0023 (0.017)	-0.0356 (0.118)
Decile 5 * high delta VIX	-0.0070 (0.010)	0.1053*** (0.035)	0.0166 (0.016)	0.2571** (0.106)	-0.0016 (0.021)	-0.1317 (0.104)	-0.0091 (0.015)	0.0198 (0.121)
Inverse Mills ratio	0.5040*** (0.012)	0.0370* (0.022)	0.4460*** (0.031)	-0.0780* (0.044)	0.3100*** (0.076)	-0.1100* (0.064)	0.0885*** (0.010)	-0.1910*** (0.061)
Observations	2,915,822	2,915,822	1,813,435	1,813,435	809,892	809,892	2,040,304	2,040,304
p_chi ²	0.000	0.102	0.000	0.070	0.000	0.086	0.000	0.002
Observed minus predicted from > 1 day	-0.132		-0.250		-0.068		-0.100	
Observed minus predicted from same day		-0.009		0.088		0.010		0.100

Table 10
Endogenous Switching Regression Model Excluding Roundtrips of *Top Ten: Volume Dealers*

The table reports the results of the two stage endogenous switching regression model. Coefficients are reported for trading activity deciles; coefficients for additional included variables from the base specification (Tables 8 and 9) are not reported for brevity. Variables are as defined in Appendix Table A1. *, **, and *** denote statistical significance at the 0.10, 0.05, and 0.01 level, respectively. Regressions include quarter fixed effects. Robust standard errors clustered by issuer are shown in parentheses.

	Stage 1: Selection Equation (Probit)				Stage 2: Spread Model							
	A & above	BBB	BB	B & below	A & above		BBB		BB		B & below	
	same day	> 1 day	same day	> 1 day	same day	> 1 day	same day	> 1 day	same day	> 1 day	same day	> 1 day
Observations	2,066,157	1,174,222	457,891	1,175,389	2,066,157	1,174,222	457,891	1,175,389				
<i>A. Deciles based on trade count</i>												
Decile 1	0.1510*** (0.023)	0.1906*** (0.031)	0.3774*** (0.033)	0.1737*** (0.035)	0.0820*** (0.015)	0.2341*** (0.037)	0.0589*** (0.020)	-0.0093 (0.069)	0.0241 (0.041)	-0.0365 (0.098)	0.0153 (0.013)	-0.1645** (0.075)
Decile 2	0.0139 (0.018)	0.0164 (0.024)	0.1528*** (0.022)	-0.0052 (0.029)	0.0056 (0.013)	0.1284*** (0.030)	-0.0226 (0.017)	-0.1197* (0.064)	-0.0260 (0.024)	-0.1561* (0.081)	-0.0339*** (0.010)	-0.2176*** (0.064)
Decile 3	-0.0213 (0.014)	-0.0389* (0.021)	0.0526*** (0.020)	-0.0475* (0.025)	-0.0197* (0.011)	0.0337 (0.026)	-0.0480*** (0.014)	-0.1681*** (0.057)	-0.0389** (0.019)	-0.1740** (0.072)	-0.0400*** (0.008)	-0.2009*** (0.059)
Decile 4	-0.0290** (0.012)	-0.0450*** (0.017)	0.0119 (0.018)	-0.0472** (0.021)	-0.0308*** (0.008)	-0.0087 (0.020)	-0.0514*** (0.012)	-0.1658*** (0.050)	-0.0379** (0.016)	-0.1317** (0.066)	-0.0276*** (0.007)	-0.1106* (0.057)
Decile 5	-0.0317*** (0.010)	-0.0416*** (0.013)	0.0054 (0.016)	-0.0465*** (0.017)	-0.0318*** (0.006)	-0.0176 (0.015)	-0.0507*** (0.010)	-0.1475*** (0.042)	-0.0345** (0.015)	-0.1068* (0.064)	-0.0185*** (0.006)	-0.1220*** (0.046)
Top ten dealer: trade count	0.1672*** (0.025)	0.0716*** (0.026)	-0.1360*** (0.041)	-0.0544 (0.039)	0.1258*** (0.009)	0.2274*** (0.018)	0.0526*** (0.016)	0.2442*** (0.018)	0.0256 (0.044)	0.2029*** (0.036)	0.0603** (0.025)	0.2046*** (0.036)
Eigenvector centrality	-19.1561*** (0.337)	-17.3917*** (0.401)	-9.2402*** (0.467)	-7.5315*** (0.313)	-5.5233*** (0.196)	-0.9627** (0.455)	-4.5576*** (0.298)	-2.3420*** (0.727)	-1.4597* (0.871)	0.7211 (0.693)	-0.7983*** (0.205)	-1.6101*** (0.523)
Dealer takes inventory (this bond)	-1.5464*** (0.030)	-1.5393*** (0.067)	-1.7153*** (0.064)	-1.7809*** (0.040)								
Dealer traded in prior year (this bond)	-0.0014*** (0.000)	-0.0021*** (0.000)	-0.0039*** (0.001)	-0.0037*** (0.000)								
Inverse Mills ratio					0.479 0.0109***	0.0545 0.0257**	0.446 0.0258***	-0.0259 0.0444	0.263 0.1510*	-0.00240 0.0762	0.137 0.0203***	-0.207 0.0458***
Observed minus predicted from > 1 day					-0.135		-0.242		-0.057		-0.089	
Observed minus predicted from same day						-0.005		0.090		0.019		0.133
<i>B. Deciles based on days traded</i>												
Decile 1	0.3600*** (0.021)	0.3854*** (0.029)	0.5131*** (0.025)	0.4579*** (0.025)	0.2051*** (0.013)	0.3257*** (0.036)	0.1626*** (0.017)	0.2241*** (0.064)	0.1022*** (0.039)	0.2095** (0.096)	0.1039*** (0.013)	0.2127*** (0.065)
Decile 2	0.1632*** (0.016)	0.1922*** (0.023)	0.2463*** (0.020)	0.1820*** (0.019)	0.0979*** (0.010)	0.2126*** (0.026)	0.0623*** (0.015)	0.0623 (0.058)	0.0326 (0.021)	0.0224 (0.066)	0.0220** (0.009)	0.0520 (0.054)
Decile 3	0.1169*** (0.013)	0.1122*** (0.019)	0.1399*** (0.015)	0.1016*** (0.016)	0.0534*** (0.009)	0.0869*** (0.023)	0.0245** (0.012)	-0.0285 (0.048)	0.0037 (0.017)	-0.0208 (0.057)	0.0002 (0.007)	0.0533 (0.050)
Decile 4	0.1112*** (0.009)	0.0970*** (0.012)	0.0930*** (0.012)	0.0751*** (0.013)	0.0340*** (0.006)	0.0297** (0.015)	0.0118 (0.010)	-0.0355 (0.044)	0.0011 (0.013)	-0.0369 (0.056)	-0.0029 (0.006)	0.0268 (0.045)
Decile 5	0.1657*** (0.008)	0.1354*** (0.011)	0.1010*** (0.012)	0.0762*** (0.010)	0.0387*** (0.003)	0.0117 (0.008)	0.0304*** (0.006)	0.0243 (0.035)	0.0089 (0.011)	-0.0278 (0.036)	0.0052 (0.003)	0.0594 (0.040)
Top ten dealer: trade count	0.1655*** (0.025)	0.0690*** (0.026)	-0.1373*** (0.040)	-0.0538 (0.039)	0.1262*** (0.010)	0.2285*** (0.018)	0.0534*** (0.016)	0.2452*** (0.018)	0.0255 (0.039)	0.2068*** (0.036)	0.0622** (0.025)	0.2102*** (0.037)
Eigenvector centrality	-19.2388*** (0.335)	-17.4740*** (0.405)	-9.2644*** (0.468)	-7.6067*** (0.310)	-5.5118*** (0.194)	-1.0147** (0.458)	-4.5506*** (0.302)	-2.4276*** (0.747)	-1.5229** (0.707)	0.7166 (0.713)	-0.8102*** (0.206)	-1.7419*** (0.525)
Dealer takes inventory (this bond)	-1.5479*** (0.029)	-1.5431*** (0.067)	-1.7179*** (0.064)	-1.7904*** (0.039)								
Dealer traded in prior year (this bond)	-0.0013*** (0.000)	-0.0019*** (0.000)	-0.0038*** (0.000)	-0.0035*** (0.000)								
Inverse Mills ratio					0.4780*** (0.011)	0.0510** (0.026)	0.4440*** (0.026)	-0.0239 (0.045)	0.273** (0.119)	0.00218 (0.076)	0.1400*** (0.020)	-0.2020*** (0.046)
Observed minus predicted from > 1 day					-0.137		-0.243		-0.060		-0.094	
Observed minus predicted from same day						-0.004		0.090		0.019		0.134